General Relativity Problem Sheet 2021: Problems and Solutions

This document records the problems and worked solutions in English. It is being written progressively from the beginning of the sheet.

Problem 1. Summation convention

Problem. For each formula below, either write the equation with all summation signs made explicit, or explain why the formula is ambiguous or ill-formed. If it is wrong, give one or more possible corrected versions.

(a) $X^a = L^a{}_b M^{bc} \hat{X}_c$.

(b) $X^a = L^b{}_c M^c{}_d \hat{X}^d$.

(c) $\delta^a{}_b = \delta^a{}_c \delta^c{}_d$.

(d) $\delta^a{}_b = \delta^a{}_b \delta^c{}_c$.

(e) $X^a = L^a{}_b \hat{X}^b + M^{ab} \hat{X}_b$.

(f) $X^a = L^a{}_b \hat{X}^b + M^a{}_c \hat{X}^c$.

(g) $X^a = L^a{}_c \hat{X}^c + M^b{}_c \hat{X}^c$.

(h) $X^a = L^a{}_c \hat{X}^c + \sum_c M^{ac} \hat{X}_c$.

(i) $R^a{}_{bac} = g^{ad} R_{bdca}$.

(j) $R_{\alpha\beta} = \partial_\rho \Gamma^\rho{}_{\beta\alpha} - \partial_\beta \Gamma^\rho{}_{\rho\alpha} + \Gamma^\rho{}_{\rho\lambda} \Gamma^\lambda{}_{\beta\alpha} - \Gamma^\xi{}_{\beta\eta} \Gamma^\eta{}_{\rho\alpha}$.

Solution. An index which is repeated exactly once upstairs and once downstairs is summed. An index occurring only once is free, and the same free indices must occur on both sides.

(a) Well formed:

$$X^a = \sum_b \sum_c L^a{}_b M^{bc} \hat{X}_c.$$

(b) Ill-formed as written: b is free on the right but not on the left, and a is free on the left but absent on the right. A possible correction is

$$X^a = L^a{}_c M^c{}_d \hat{X}^d = \sum_c \sum_d L^a{}_c M^c{}_d \hat{X}^d.$$

(c) Ill-formed: the right-hand side has the free index d rather than b. A corrected identity is

$$\delta^a{}_b = \delta^a{}_c \delta^c{}_b = \sum_c \delta^a{}_c \delta^c{}_b.$$

(d) Formally well formed, but generally false. Since $\delta^c{}_c = \sum_c 1 = n$, the right-hand side is $n\delta^a{}_b$. A correct version is simply $\delta^a{}_b = \delta^a{}_c \delta^c{}_b$.

(e) Well formed:
$$X^a = \sum_b L^a{}_b \hat{X}^b + \sum_b M^{ab} \hat{X}_b.$$

(f) Well formed:
$$X^a = \sum_b L^a{}_b \hat{X}^b + \sum_c M^a{}_c \hat{X}^c.$$

The two dummy indices could both be renamed b.

(g) Ill-formed: the second term has free index b, not a. One possible correction is
$$X^a = L^a{}_c \hat{X}^c + M^a{}_c \hat{X}^c.$$

(h) Well formed because the summation over c is written explicitly in the second term:
$$X^a = \sum_c L^a{}_c \hat{X}^c + \sum_c M^{ac} \hat{X}_c.$$

(i) Ill-formed as a contraction formula for $R^a{}_{bac}$: a is both free and summed on the left. A possible intended formula is
$$R^a{}_{bcd} = g^{ae} R_{ebcd}.$$

If the intended object was a Ricci-type contraction, one should use a different dummy index, for example $R_{bc} = R^a{}_{bac} = g^{ad} R_{dbac}$, subject to the author's curvature sign convention.

(j) Well formed:
$$R_{\alpha\beta} = \sum_\rho \partial_\rho \Gamma^\rho{}_{\beta\alpha} - \sum_\rho \partial_\beta \Gamma^\rho{}_{\rho\alpha} + \sum_{\rho,\lambda} \Gamma^\rho{}_{\rho\lambda} \Gamma^\lambda{}_{\beta\alpha} - \sum_{\xi,\eta,\rho} \Gamma^\xi{}_{\beta\eta} \Gamma^\eta{}_{\rho\alpha}.$$

However, the last term has ρ appearing only once in $\Gamma^\eta{}_{\rho\alpha}$ after summing over ξ, η, so the displayed formula is actually ill-formed. A standard corrected version is
$$R_{\alpha\beta} = \partial_\rho \Gamma^\rho{}_{\beta\alpha} - \partial_\beta \Gamma^\rho{}_{\rho\alpha} + \Gamma^\rho{}_{\rho\lambda} \Gamma^\lambda{}_{\beta\alpha} - \Gamma^\rho{}_{\beta\lambda} \Gamma^\lambda{}_{\rho\alpha}.$$

Problem 2. Symmetrisation and antisymmetrisation

Problem. With brackets denoting complete antisymmetrisation and parentheses denoting complete symmetrisation, prove the following identities:
$$A_{(\mu\nu)} = A_{(\nu\mu)}, \quad A_{[\mu\nu]} = -A_{[\nu\mu]},$$
$$A_{\mu\nu} = A_{[\mu\nu]} + A_{(\mu\nu)},$$
$$A_{[\mu\nu]} B^{\mu\nu} = A_{\mu\nu} B^{[\mu\nu]} = A_{[\mu\nu]} B^{[\mu\nu]},$$
$$A_{(\mu\nu)} B^{\mu\nu} = A_{\mu\nu} B^{(\mu\nu)} = A_{(\mu\nu)} B^{(\mu\nu)},$$

and the analogous statements listed on the sheet for symmetric and antisymmetric tensors, three-index antisymmetrisation, and antisymmetrised Kronecker deltas.

Solution. For two indices the definitions give
$$A_{(\mu\nu)} = \frac{1}{2}(A_{\mu\nu} + A_{\nu\mu}), \qquad A_{[\mu\nu]} = \frac{1}{2}(A_{\mu\nu} - A_{\nu\mu}).$$

Interchanging μ and ν proves $A_{(\mu\nu)} = A_{(\nu\mu)}$ and $A_{[\mu\nu]} = -A_{[\nu\mu]}$. Adding the two definitions gives
$$A_{(\mu\nu)} + A_{[\mu\nu]} = A_{\mu\nu}.$$

For contractions, dummy indices may be renamed:

$$A_{[\mu\nu]}B^{\mu\nu} = \frac{1}{2}(A_{\mu\nu} - A_{\nu\mu})B^{\mu\nu}$$
$$= \frac{1}{2}A_{\mu\nu}B^{\mu\nu} - \frac{1}{2}A_{\mu\nu}B^{\nu\mu} = A_{\mu\nu}B^{[\mu\nu]}.$$

Replacing B by $B_{[\cdot\cdot]} + B_{(\cdot\cdot)}$, the mixed contraction of an antisymmetric tensor with a symmetric tensor vanishes, so

$$A_{[\mu\nu]}B^{\mu\nu} = A_{[\mu\nu]}B^{[\mu\nu]}.$$

The symmetric version is identical with brackets replaced by parentheses:

$$A_{(\mu\nu)}B^{\mu\nu} = A_{\mu\nu}B^{(\mu\nu)} = A_{(\mu\nu)}B^{(\mu\nu)}.$$

If $C_{\alpha\beta} = C_{\beta\alpha}$ and $D^{\alpha\beta} = -D^{\beta\alpha}$, then

$$C_{\alpha\beta}D^{\alpha\beta} = C_{\beta\alpha}D^{\beta\alpha} = -C_{\alpha\beta}D^{\alpha\beta},$$

hence the contraction is zero. For three indices the same averaging argument over the permutation group gives

$$A_{[\mu\nu\rho]}B^{\mu\nu\rho} = A_{\mu\nu\rho}B^{[\mu\nu\rho]},$$

because after renaming dummy indices each permutation in the antisymmetriser can be moved from A to B with the same sign.

Finally,

$$\delta^{[\alpha}{}_{\mu}\delta^{\gamma]}{}_{\rho} = \frac{1}{2}(\delta^{\alpha}{}_{\mu}\delta^{\gamma}{}_{\rho} - \delta^{\gamma}{}_{\mu}\delta^{\alpha}{}_{\rho}) = \delta^{\alpha}{}_{[\mu}\delta^{\gamma}{}_{\rho]} = \delta^{[\alpha}{}_{[\mu}\delta^{\gamma]}{}_{\rho]}.$$

The three-delta identity follows in the same way by expanding the six terms of the complete antisymmetrisation:

$$\delta^{[\alpha}{}_{\mu}\delta^{\beta}{}_{\nu}\delta^{\gamma]}{}_{\rho} = \delta^{\alpha}{}_{[\mu}\delta^{\beta}{}_{\nu}\delta^{\gamma}{}_{\rho]} = \delta^{[\alpha}{}_{[\mu}\delta^{\beta}{}_{\nu}\delta^{\gamma]}{}_{\rho]}.$$

Both sides are the determinant of the 3×3 matrix $(\delta^{\alpha_i}{}_{\mu_j})$, divided by 3!.

Problem 3. Changes of coordinates

Problem. In Minkowski space with inertial coordinates (t, x, y, z) and metric

$$g = -\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2,$$

let $X = (1, 1, 0, 0)^T$ and let $\alpha = (1, 1, 0, 0)$ in these inertial coordinates. Find the metric coefficients and the components of X and α in:

(a) $\tilde{x}^0 = t - z,\ \tilde{x}^1 = r,\ \tilde{x}^2 = \theta,\ \tilde{x}^3 = z$, with $x = r\cos\theta,\ y = r\sin\theta$;

(b) $\bar{x}^0 = \tau,\ \bar{x}^1 = \varphi,\ \bar{x}^2 = y,\ \bar{x}^3 = z$, with $t = \tau\cosh\varphi,\ x = \tau\sinh\varphi$;

(c) $\hat{x}^0 = t,\ \hat{x}^1 = r,\ \hat{x}^2 = \theta,\ \hat{x}^3 = \varphi$, with spherical polar spatial coordinates.

State the region covered in each case.

Solution.

(a) Put $u = t - z$, so $t = u + z$. Also

$$\mathrm{d}x = \cos\theta\,\mathrm{d}r - r\sin\theta\,\mathrm{d}\theta, \qquad \mathrm{d}y = \sin\theta\,\mathrm{d}r + r\cos\theta\,\mathrm{d}\theta.$$

Thus

$$g = -\mathrm{d}u^2 - 2\,\mathrm{d}u\,\mathrm{d}z + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2.$$

In coordinates (u, r, θ, z),

$$(\tilde{g}_{ab}) = \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

For vectors, $\tilde{X}^a = (\partial \tilde{x}^a / \partial x^b) X^b$, hence

$$\tilde{X} = (1, \cos\theta, -\sin\theta/r, 0).$$

For one-forms, $\alpha = \mathrm{d}t + \mathrm{d}x = \mathrm{d}u + \mathrm{d}z + \cos\theta\, \mathrm{d}r - r\sin\theta\, \mathrm{d}\theta$, so

$$\tilde{\alpha} = (1, \cos\theta, -r\sin\theta, 1).$$

The chart covers $r > 0$, all $u, z \in \mathbb{R}$, and θ modulo 2π; as a single chart one takes an interval of length 2π, excluding the z-axis.

(b) Since

$$\mathrm{d}t = \cosh\varphi\, \mathrm{d}\tau + \tau \sinh\varphi\, \mathrm{d}\varphi, \quad \mathrm{d}x = \sinh\varphi\, \mathrm{d}\tau + \tau\cosh\varphi\, \mathrm{d}\varphi,$$

we get

$$g = -\mathrm{d}\tau^2 + \tau^2 \mathrm{d}\varphi^2 + \mathrm{d}y^2 + \mathrm{d}z^2.$$

Thus

$$(\bar{g}_{ab}) = \mathrm{diag}(-1, \tau^2, 1, 1).$$

The vector components are

$$\bar{X}^\tau = \partial_t \tau + \partial_x \tau = \cosh\varphi - \sinh\varphi = e^{-\varphi},$$

$$\bar{X}^\varphi = \partial_t \varphi + \partial_x \varphi = -\frac{\sinh\varphi}{\tau} + \frac{\cosh\varphi}{\tau} = \frac{e^{-\varphi}}{\tau}, \qquad \bar{X}^y = \bar{X}^z = 0.$$

The one-form is

$$\alpha = \mathrm{d}t + \mathrm{d}x = e^\varphi \mathrm{d}\tau + \tau e^\varphi \mathrm{d}\varphi,$$

so

$$\bar{\alpha} = (e^\varphi, \tau e^\varphi, 0, 0).$$

For $\tau > 0$ this covers the right Rindler wedge $x > |t|$. Allowing $\tau < 0$ covers the left wedge as well, but the usual Rindler chart takes $\tau > 0$.

(c) Spherical coordinates give

$$\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = \mathrm{d}r^2 + r^2 \mathrm{d}\theta^2 + r^2 \sin^2\theta\, \mathrm{d}\varphi^2,$$

so

$$(\hat{g}_{ab}) = \mathrm{diag}(-1, 1, r^2, r^2 \sin^2\theta).$$

Using

$$\partial_x r = \sin\theta\cos\varphi, \quad \partial_x \theta = \frac{\cos\theta\cos\varphi}{r}, \quad \partial_x \varphi = -\frac{\sin\varphi}{r\sin\theta},$$

we find

$$\hat{X} = (1, \sin\theta\cos\varphi, \cos\theta\cos\varphi/r, -\sin\varphi/(r\sin\theta)).$$

Also

$$\alpha = \mathrm{d}t + \mathrm{d}x = \mathrm{d}t + \sin\theta\cos\varphi\, \mathrm{d}r + r\cos\theta\cos\varphi\, \mathrm{d}\theta - r\sin\theta\sin\varphi\, \mathrm{d}\varphi,$$

hence

$$\hat{\alpha} = (1, \sin\theta\cos\varphi, r\cos\theta\cos\varphi, -r\sin\theta\sin\varphi).$$

The chart covers $r > 0$ away from the polar axis, with the usual angular coordinate restrictions.

Problem 4. Stereographic projection

Problem. Let $(x, y) \in \mathbb{R}^2$ be stereographic coordinates on the unit two-sphere, obtained by projecting from the north pole to the plane tangent at the south pole. Express

$$g = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2$$

in these coordinates.

Solution. For projection from the north pole to the plane tangent at the south pole, the radial coordinate $\rho = \sqrt{x^2 + y^2}$ satisfies

$$\rho = 2\cot\frac{\theta}{2}.$$

Equivalently, if $R = \rho/2$, then

$$\sin\theta = \frac{2R}{1+R^2} = \frac{4\rho}{\rho^2+4}, \qquad \mathrm{d}\theta = -\frac{4}{\rho^2+4}\,\mathrm{d}\rho.$$

Since $\mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}\rho^2 + \rho^2\mathrm{d}\phi^2$, the metric becomes

$$g = \frac{16}{(\rho^2+4)^2}\left(\mathrm{d}\rho^2 + \rho^2\mathrm{d}\phi^2\right) = \frac{16}{(x^2+y^2+4)^2}\left(\mathrm{d}x^2 + \mathrm{d}y^2\right).$$

If one instead uses projection to the equatorial plane, the familiar conformal factor is $4/(1 + x^2 + y^2)^2$; the difference is only the scale of the planar coordinates.

Problem 5. Coordinates on a manifold

Problem.

(a) Find an atlas and transition functions for the circle S^1.

(b) The infinite cylinder $x^2 + y^2 = 1$ with $z \in \mathbb{R}$ is mapped to $\mathbb{R}^2 \setminus \{0\}$ by

$$(x, y, z) \mapsto (X, Y) = (xe^z, ye^z).$$

Find the inverse map and describe the images of the listed curves.

Solution.

(a) A standard two-chart atlas is obtained by stereographic projection from the north and south poles. Write

$$S^1 = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 = 1\}.$$

Let $N = (0, 1)$, $S = (0, -1)$. Define

$$\phi_N : S^1 \setminus \{N\} \to \mathbb{R}, \qquad \phi_N(u, v) = \frac{u}{1-v},$$

$$\phi_S : S^1 \setminus \{S\} \to \mathbb{R}, \qquad \phi_S(u, v) = \frac{u}{1+v}.$$

On the overlap $S^1 \setminus \{N, S\}$,

$$\phi_S \circ \phi_N^{-1}(s) = \frac{1}{s}, \qquad \phi_N \circ \phi_S^{-1}(s) = \frac{1}{s}.$$

Thus the transition functions are smooth on $\mathbb{R} \setminus \{0\}$.

(b) Since $X = xe^z,\ Y = ye^z$ and $x^2 + y^2 = 1$,

$$e^z = \sqrt{X^2 + Y^2}, \qquad z = \log\sqrt{X^2 + Y^2},$$

and

$$x = \frac{X}{\sqrt{X^2 + Y^2}}, \qquad y = \frac{Y}{\sqrt{X^2 + Y^2}}.$$

Thus the inverse is

$$(X, Y) \mapsto \left(\frac{X}{\sqrt{X^2 + Y^2}}, \frac{Y}{\sqrt{X^2 + Y^2}}, \log\sqrt{X^2 + Y^2}\right).$$

The line $(x, y, z) = (1, 0, z)$ maps to the positive X-axis, $(X, Y) = (e^z, 0)$. The circles $z = c$ map to Euclidean circles

$$X^2 + Y^2 = e^{2c},$$

so $z = 1, -1, 0$ give radii $e, e^{-1}, 1$. The spiral $(\cos a, \sin a, a)$ maps to

$$(X, Y) = e^a(\cos a, \sin a),$$

which is a logarithmic spiral.

Problem 6. Vector as a differential operator

Problem. Show that (X^i) is a vector field, in the coordinate-transformation sense, if and only if for every scalar field f, the expression $X^i\partial_i f$ is a scalar.

Solution. Assume first that X transforms as a vector:

$$\tilde{X}^i(\tilde{x}) = \frac{\partial\tilde{x}^i}{\partial x^j}X^j(x).$$

The chain rule gives

$$\frac{\partial\tilde{f}}{\partial\tilde{x}^i} = \frac{\partial x^k}{\partial\tilde{x}^i}\frac{\partial f}{\partial x^k}.$$

Therefore

$$\tilde{X}^i\tilde{\partial}_i\tilde{f} = \frac{\partial\tilde{x}^i}{\partial x^j}X^j\frac{\partial x^k}{\partial\tilde{x}^i}\partial_k f = X^j\partial_j f,$$

so $X^i\partial_i f$ is scalar.

Conversely, suppose $X^i\partial_i f$ is scalar for all f. In the new coordinates choose the scalar functions $f = \tilde{x}^a$. Then

$$\tilde{X}^a = \tilde{X}^i\tilde{\partial}_i\tilde{x}^a = X^j\partial_j\tilde{x}^a = \frac{\partial\tilde{x}^a}{\partial x^j}X^j.$$

This is precisely the vector transformation law.

Problem 7. Lie bracket

Problem. For vector fields X, Y, define

$$[X, Y](f) = X(Y(f)) - Y(X(f)).$$

Show that in local coordinates

$$[X, Y]^i = X^j\partial_j Y^i - Y^j\partial_j X^i,$$

that $[X, Y]$ is a vector field, and that the Jacobi identity holds.

Solution. Write $X = X^i\partial_i$ and $Y = Y^i\partial_i$. Then

$$X(Y(f)) = X^j\partial_j(Y^i\partial_i f) = X^j(\partial_j Y^i)\partial_i f + X^j Y^i \partial_j\partial_i f,$$
$$Y(X(f)) = Y^j(\partial_j X^i)\partial_i f + Y^j X^i \partial_j\partial_i f.$$

The second-derivative terms cancel because partial derivatives commute and the coefficients are symmetric after relabelling dummy indices. Thus

$$[X, Y] = \big(X^j\partial_j Y^i - Y^j\partial_j X^i\big)\partial_i.$$

Since the commutator sends every scalar f to a scalar and is a homogeneous linear first-order differential operator, Problem 6 shows that it is a vector field.

For the Jacobi identity, use commutators of operators. For any scalar f,

$$[X, [Y, Z]]f + [Y, [Z, X]]f + [Z, [X, Y]]f = 0$$

after expanding all twelve terms:

$$XYZf - XZYf - YZXf + ZYXf + YZXf - YXZf - ZXYf + XYZf + ZXYf - ZYXf - XYZf + YXZf = 0.$$

Therefore

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Problem 8. Raising and lowering of indices

Problem. Let $g_{\mu\nu}$ be symmetric and non-degenerate. Define

$$B_\alpha = g_{\alpha\beta}A^\beta, \qquad C^\gamma = g^{\gamma\sigma}B_\sigma.$$

Show that $C^\gamma = A^\gamma$. Explain the meaning for raising and lowering indices. Also show that $A_\alpha B^\alpha = A^\alpha B_\alpha$.

Solution. Since $g^{\gamma\sigma}$ is the inverse matrix of $g_{\alpha\beta}$,

$$C^\gamma = g^{\gamma\sigma}B_\sigma = g^{\gamma\sigma}g_{\sigma\beta}A^\beta = \delta^\gamma{}_\beta A^\beta = A^\gamma.$$

Thus lowering an index with $g_{\alpha\beta}$ and then raising it with $g^{\alpha\beta}$ returns the original vector; the two operations are inverse linear isomorphisms between vectors and covectors.

Finally,

$$A_\alpha B^\alpha = g_{\alpha\beta}A^\beta B^\alpha = g_{\beta\alpha}A^\alpha B^\beta = A^\alpha g_{\alpha\beta}B^\beta = A^\alpha B_\alpha,$$

where symmetry of g and dummy-index relabelling were used.

Problem 9. Killing vectors

Problem.

(a) Suppose X has the property that for every affinely parametrised geodesic γ, the quantity $g(X, \dot\gamma)$ is constant along γ. Show that X satisfies the Killing equation

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = 0.$$

(b) Show that constant linear combinations of Killing vector fields are again Killing.

(c) Let $x^\mu(s)$ be an integral curve of a Killing field X, so $\dot{x}^\mu = X^\mu(x(s))$. Show that the integral curves are geodesics if and only if

$$\nabla_\alpha(g(X,X)) = 0$$

along the curve.

Solution.

(a) Along an affinely parametrised geodesic with tangent u^μ,

$$\frac{\mathrm{d}}{\mathrm{d}s} g(X,u) = u^\mu \nabla_\mu (X_\nu u^\nu) = u^\mu u^\nu \nabla_\mu X_\nu + X_\nu u^\mu \nabla_\mu u^\nu.$$

The second term vanishes by the geodesic equation. Since the expression is assumed to vanish for every geodesic and every possible initial tangent u,

$$u^\mu u^\nu \nabla_\mu X_\nu = 0$$

for all u. Only the symmetric part contributes to this quadratic form, so

$$u^\mu u^\nu \nabla_{(\mu} X_{\nu)} = 0 \quad \text{for all } u.$$

By polarization, $\nabla_{(\mu} X_{\nu)} = 0$, which is exactly

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = 0.$$

(b) If X, Y are Killing and $a, b \in \mathbb{R}$ are constant, then

$$\nabla_\mu(aX_\nu + bY_\nu) + \nabla_\nu(aX_\mu + bY_\mu) = a(\nabla_\mu X_\nu + \nabla_\nu X_\mu) + b(\nabla_\mu Y_\nu + \nabla_\nu Y_\mu) = 0.$$

Thus the Killing fields form a vector space.

(c) An integral curve of X has acceleration

$$X^\nu \nabla_\nu X^\mu.$$

Lowering the index and using the Killing equation,

$$X^\nu \nabla_\nu X_\mu = -X^\nu \nabla_\mu X_\nu = -\frac{1}{2}\nabla_\mu (X^\nu X_\nu).$$

Raising the index gives

$$X^\nu \nabla_\nu X^\mu = -\frac{1}{2}\nabla^\mu g(X,X).$$

Hence the integral curve is an affinely parametrised geodesic precisely when this acceleration vanishes, i.e. precisely when $\nabla_\alpha(g(X,X)) = 0$ along the curve.

Problem 10. Christoffel symbols

Problem.

(a) Compute the Christoffel symbols for the Euclidean metric $g = \mathrm{d}\rho^2 + \rho^2 \mathrm{d}\phi^2$ and for the round metric $h = \mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2$.

(b) Show that the Euler–Lagrange equations for

$$L = \frac{1}{2} g_{ab} \dot{x}^a \dot{x}^b$$

are the geodesic equations.

(c) Show that if g is independent of x^1, then $g(\dot{x}, \partial_1)$ is conserved along geodesics.

(d) For

$$g = -e^{2f(r)}\mathrm{d}t^2 + e^{-2f(r)}\mathrm{d}r^2 + r^2\mathrm{d}\phi^2,$$

write the geodesic equations and the obvious constants of motion.

(e) Compute the Christoffel symbols for this metric.

(f) Derive the coordinate transformation law for Christoffel symbols and explain why they are not tensor components.

Solution.

(a) For a metric

$$\mathrm{d}x^2 + e^{2F(x)}\mathrm{d}y^2,$$

the only non-zero Christoffel symbols are

$$\Gamma^x{}_{yy} = -F'e^{2F}, \qquad \Gamma^y{}_{xy} = \Gamma^y{}_{yx} = F'.$$

For the Euclidean polar metric, $x = \rho$, $y = \phi$, and $e^{2F} = \rho^2$, so $F'(\rho) = 1/\rho$. Hence

$$\Gamma^\rho{}_{\phi\phi} = -\rho, \qquad \Gamma^\phi{}_{\rho\phi} = \Gamma^\phi{}_{\phi\rho} = \frac{1}{\rho}.$$

For the unit sphere, $x = \theta$, $y = \phi$, and $e^{2F} = \sin^2\theta$, so $F' = \cot\theta$. Hence

$$\Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^\phi{}_{\theta\phi} = \Gamma^\phi{}_{\phi\theta} = \cot\theta.$$

(b) The Euler–Lagrange equations are

$$\frac{\mathrm{d}}{\mathrm{d}s}(g_{ab}\dot{x}^b) = \frac{1}{2}\partial_a g_{bc}\dot{x}^b\dot{x}^c.$$

Expanding the left-hand side gives

$$g_{ab}\ddot{x}^b + \partial_c g_{ab}\dot{x}^c\dot{x}^b - \frac{1}{2}\partial_a g_{bc}\dot{x}^b\dot{x}^c = 0.$$

Multiplying by g^{da} and symmetrising the terms multiplying $\dot{x}^b\dot{x}^c$,

$$\ddot{x}^d + \frac{1}{2}g^{da}(\partial_b g_{ac} + \partial_c g_{ab} - \partial_a g_{bc})\dot{x}^b\dot{x}^c = 0.$$

The coefficient is $\Gamma^d{}_{bc}$, so this is the geodesic equation.

(c) If g_{ab} is independent of x^1, then $\partial L/\partial x^1 = 0$. The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}s}\frac{\partial L}{\partial \dot{x}^1} = 0.$$

But

$$\frac{\partial L}{\partial \dot{x}^1} = g_{1b}\dot{x}^b = g(\partial_1, \dot{x}),$$

so this quantity is conserved.

(d) The Lagrangian is
$$L = \frac{1}{2}\left(-e^{2f}\dot{t}^2 + e^{-2f}\dot{r}^2 + r^2\dot{\phi}^2\right).$$
Since t and ϕ are cyclic,
$$E = e^{2f}\dot{t}, \qquad J = r^2\dot{\phi}$$
are conserved. The radial Euler–Lagrange equation is
$$\frac{\mathrm{d}}{\mathrm{d}s}(e^{-2f}\dot{r}) = -f'e^{2f}\dot{t}^2 - f'e^{-2f}\dot{r}^2 + r\dot{\phi}^2,$$
or equivalently
$$\ddot{r} - f'\dot{r}^2 + f'e^{4f}\dot{t}^2 - re^{2f}\dot{\phi}^2 = 0.$$
Together with
$$\ddot{t} + 2f'\dot{t}\dot{r} = 0, \qquad \ddot{\phi} + \frac{2}{r}\dot{r}\dot{\phi} = 0,$$
these are the geodesic equations.

(e) In coordinates (t, r, ϕ), the non-zero Christoffel symbols are
$$\Gamma^t{}_{tr} = \Gamma^t{}_{rt} = f',$$
$$\Gamma^r{}_{tt} = f'e^{4f}, \qquad \Gamma^r{}_{rr} = -f', \qquad \Gamma^r{}_{\phi\phi} = -re^{2f},$$
$$\Gamma^\phi{}_{r\phi} = \Gamma^\phi{}_{\phi r} = \frac{1}{r}.$$
Substituting these into $\ddot{x}^a + \Gamma^a{}_{bc}\dot{x}^b\dot{x}^c = 0$ reproduces the equations in part (d).

(f) Let $x^a = x^a(\tilde{x})$. A vector V satisfies
$$\nabla_b V^a = \partial_b V^a + \Gamma^a{}_{bc}V^c.$$
Writing this equation in both coordinate systems and applying the chain rule to
$$V^a = \frac{\partial x^a}{\partial \tilde{x}^d}\tilde{V}^d$$
gives
$$\Gamma^a{}_{bc} = \tilde{\Gamma}^d{}_{ef}\frac{\partial x^a}{\partial \tilde{x}^d}\frac{\partial \tilde{x}^e}{\partial x^b}\frac{\partial \tilde{x}^f}{\partial x^c} + \frac{\partial x^a}{\partial \tilde{x}^d}\frac{\partial^2 \tilde{x}^d}{\partial x^b \partial x^c}.$$
The second-derivative term is inhomogeneous. A tensor transformation law is linear and homogeneous in the old components, so the Christoffel symbols are not the components of a tensor.

Problem 11. Some tensor manipulations

Problem.

(a) Show that for all functions f,
$$\nabla_a\nabla_b f = \nabla_b\nabla_a f$$
if and only if ∇ is torsion-free.

(b) With $\nabla f = g^{ab}\partial_a f\,\partial_b$, explain $\mathrm{d}f(\nabla f)$ and $\nabla f(f)$, and prove
$$g(\nabla f, \nabla f) = g^{ab}\partial_a f\,\partial_b f = \mathrm{d}f(\nabla f) = \nabla f(f).$$
Conclude how to read the causal character of ∇x^α from the inverse metric.

Solution.

(a) For a scalar f, $\nabla_b f = \partial_b f$. Therefore

$$\nabla_a \nabla_b f = \partial_a \partial_b f - \Gamma^c{}_{ba} \partial_c f.$$

Hence

$$\nabla_a \nabla_b f - \nabla_b \nabla_a f = \big(\Gamma^c{}_{ab} - \Gamma^c{}_{ba}\big)\partial_c f = T^c{}_{ab} \partial_c f,$$

where $T^c{}_{ab}$ is the torsion tensor in coordinates. If the connection is torsion-free this vanishes for every f. Conversely, if it vanishes for every f, choose $f = x^c$ locally; then $T^c{}_{ab} = 0$, so the connection is torsion-free.

(b) The expression $\mathrm{d}f(\nabla f)$ means the one-form $\mathrm{d}f$ applied to the vector ∇f. The expression $\nabla f(f)$ means the directional derivative of f along the vector field ∇f. Since

$$(\nabla f)^a = g^{ab} \partial_b f,$$

we have

$$g(\nabla f, \nabla f) = g_{ab} g^{ac} \partial_c f \, g^{bd} \partial_d f = g^{cd} \partial_c f \, \partial_d f.$$

Also,

$$\mathrm{d}f(\nabla f) = \partial_a f (\nabla f)^a = g^{ab} \partial_a f \, \partial_b f,$$

and the same calculation gives

$$\nabla f(f) = (\nabla f)^a \partial_a f = g^{ab} \partial_a f \, \partial_b f.$$

For $f = x^\alpha$, $\partial_a f = \delta^\alpha{}_a$, so

$$g(\nabla x^\alpha, \nabla x^\alpha) = g^{\alpha\alpha}$$

with no summation over α. Thus the sign of the inverse metric component $g^{\alpha\alpha}$ determines whether the gradient of the coordinate function is timelike, spacelike, or null, according to the chosen signature convention.

Problem 12. Symmetries of the curvature tensor

Problem. State what it means for a connection to be metric and torsion-free. Assuming torsion-freeness, prove the exterior-derivative identities for covectors and two-forms, derive the curvature action on covectors and tensors, and establish the usual algebraic symmetries of the Riemann tensor for the Levi–Civita connection.

Solution.

(a) A connection is metric if

$$\nabla_a g_{bc} = 0.$$

It is torsion-free if

$$T^c{}_{ab} := \Gamma^c{}_{ab} - \Gamma^c{}_{ba} = 0,$$

or, invariantly, $\nabla_X Y - \nabla_Y X = [X, Y]$.

(b) For a covector A_b,

$$\nabla_a A_b - \nabla_b A_a = \partial_a A_b - \partial_b A_a - (\Gamma^c{}_{ba} - \Gamma^c{}_{ab}) A_c.$$

The last term vanishes for a torsion-free connection, so

$$H_{ab} := \nabla_a A_b - \nabla_b A_a = \partial_a A_b - \partial_b A_a$$

is independent of the connection. For an antisymmetric tensor F_{ab},

$$\nabla_a F_{bc} + \nabla_b F_{ca} + \nabla_c F_{ab} = \partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab},$$

because the Christoffel terms cancel in pairs using $\Gamma^d{}_{ab} = \Gamma^d{}_{ba}$ and $F_{ab} = -F_{ba}$. Thus this cyclic combination is also connection-independent.

(c) Since $H = \mathrm{d}A$ as a two-form, the identity is $\mathrm{d}H = \mathrm{d}^2 A = 0$. In indices,

$$\nabla_a H_{bc} + \nabla_b H_{ca} + \nabla_c H_{ab} = 0.$$

(d) If the curvature on vectors is defined by

$$(\nabla_a \nabla_b - \nabla_b \nabla_a) V^c = R^c{}_{dab} V^d,$$

then its action on covectors is the dual action:

$$(\nabla_a \nabla_b - \nabla_b \nabla_a) A_c = -R^d{}_{cab} A_d.$$

This follows by applying the commutator to the scalar $A_c V^c$, whose curvature commutator is zero.

(e) Apply part (d) to $H_{bc} = \nabla_b A_c - \nabla_c A_b$, then use the cyclic identity in part (c). Since A_d is arbitrary, one obtains the first Bianchi identity

$$R^d{}_{abc} + R^d{}_{bca} + R^d{}_{cab} = 0.$$

With lowered indices this is $R_{dabc} + R_{dbca} + R_{dcab} = 0$, up to the chosen placement convention for the curvature indices.

(f) For a $(0, 2)$-tensor T_{cd}, curvature acts on each covariant index:

$$(\nabla_a \nabla_b - \nabla_b \nabla_a) T_{cd} = -R^e{}_{cab} T_{ed} - R^e{}_{dab} T_{ce}.$$

This follows either by applying the previous covector result to each slot or by using the Leibniz rule on decomposable tensors.

(g) If the connection is metric, the commutator applied to g_{cd} vanishes:

$$0 = (\nabla_a \nabla_b - \nabla_b \nabla_a) g_{cd} = -R^e{}_{cab} g_{ed} - R^e{}_{dab} g_{ce}.$$

Lowering indices gives

$$R_{dcab} + R_{cdab} = 0,$$

which is the antisymmetry in the first pair:

$$R_{abcd} = -R_{bacd}.$$

The antisymmetry in the last pair,

$$R_{abcd} = -R_{abdc},$$

is already present from the definition as a commutator in the two derivative indices.

(h) Assume

$$R_{abcd} = R_{[ab]cd} = R_{ab[cd]}, \qquad R_{a[bcd]} = 0.$$

Expanding the Bianchi identity and writing down the same identity with cyclically permuted pairs gives the standard algebraic cancellation:

$$R_{abcd} + R_{acdb} + R_{adbc} = 0,$$

$$R_{cdab} + R_{cadb} + R_{cbad} = 0.$$

Using antisymmetry in each pair to rewrite the last two terms in both equations and subtracting, all terms cancel except $R_{abcd} - R_{cdab}$. Hence

$$R_{abcd} = R_{cdab}.$$

Problem 13. Covariant derivative

Problem.

(a) For

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z,$$

show that $R(fX,Y)Z = R(X,fY)Z = R(X,Y)(fZ) = fR(X,Y)Z$.

(b) Extend ∇ to two-contravariant tensors and show

$$\nabla_k T^{ij} = \partial_k T^{ij} + \Gamma^i{}_{lk} T^{lj} + \Gamma^j{}_{lk} T^{il}.$$

(c) Define

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y)$$

and show

$$\nabla_k \alpha_i = \partial_k \alpha_i - \Gamma^l{}_{ik} \alpha_l.$$

Solution.

(a) Use $\nabla_{fX} = f\nabla_X$ and

$$[fX,Y] = f[X,Y] - Y(f)X.$$

Then

$$\begin{aligned} R(fX,Y)Z &= f\nabla_X \nabla_Y Z - \nabla_Y (f\nabla_X Z) - \nabla_{f[X,Y]-Y(f)X} Z \\ &= f\nabla_X \nabla_Y Z - f\nabla_Y \nabla_X Z - Y(f)\nabla_X Z - f\nabla_{[X,Y]} Z + Y(f)\nabla_X Z \\ &= fR(X,Y)Z. \end{aligned}$$

The proof for $R(X,fY)Z$ is analogous. Finally,

$$\nabla_Y(fZ) = Y(f)Z + f\nabla_Y Z$$

and the corresponding derivative terms cancel, giving

$$R(X,Y)(fZ) = fR(X,Y)Z.$$

Thus curvature is tensorial in all three arguments.

(b) Define

$$\nabla_X(Y \otimes Z) = (\nabla_X Y) \otimes Z + Y \otimes (\nabla_X Z)$$

and extend by linearity and the Leibniz rule. For $T = T^{ij}\partial_i \otimes \partial_j$,

$$\nabla_k T = \partial_k T^{ij} \partial_i \otimes \partial_j + T^{ij} \nabla_k \partial_i \otimes \partial_j + T^{ij} \partial_i \otimes \nabla_k \partial_j.$$

Since $\nabla_k \partial_i = \Gamma^l{}_{ik} \partial_l$, relabelling dummy indices gives

$$\nabla_k T^{ij} = \partial_k T^{ij} + \Gamma^i{}_{lk} T^{lj} + \Gamma^j{}_{lk} T^{il}.$$

(c) Put $X = \partial_k$ and $Y = \partial_i$. Then

$$(\nabla_k \alpha)(\partial_i) = \partial_k(\alpha_i) - \alpha(\nabla_k \partial_i) = \partial_k \alpha_i - \alpha(\Gamma^l{}_{ik} \partial_l).$$

Therefore

$$\nabla_k \alpha_i = \partial_k \alpha_i - \Gamma^l{}_{ik} \alpha_l.$$

Problem 14. Counting components

Problem.

(a) In four dimensions, show that a totally antisymmetric tensor $T_{abcde} = T_{[abcde]}$ vanishes.

(b) Count the independent components in n dimensions of: antisymmetric two-tensors; totally antisymmetric k-tensors; tensors $R_{abcd} = R_{[ab]cd} = R_{ab[cd]}$; tensors with additionally $R_{abcd} = R_{cdab}$.

(c) Show that a tensor with the symmetries of the Riemann tensor has 20 independent components in four dimensions.

Solution.

(a) In four dimensions, five indices chosen from $\{1, 2, 3, 4\}$ must repeat at least one value. A totally antisymmetric tensor changes sign when the two equal slots are exchanged, but the component is unchanged; hence it is equal to its negative and is zero.

(b) An antisymmetric two-tensor is determined by choosing an unordered pair of distinct indices:
$$\binom{n}{2} = \frac{n(n-1)}{2}.$$
A totally antisymmetric k-tensor has
$$\binom{n}{k}$$
independent components for $k \le n$, and no independent components for $k > n$.

For $R_{abcd} = R_{[ab]cd} = R_{ab[cd]}$, each antisymmetric pair behaves like one index taking
$$N = \binom{n}{2} = \frac{n(n-1)}{2}$$
possible values. With no pair-exchange symmetry, the count is
$$N^2 = \left(\frac{n(n-1)}{2}\right)^2.$$
If also $R_{abcd} = R_{cdab}$, then the tensor is symmetric as a bilinear form on the N-dimensional space of two-forms, so the count is
$$\frac{N(N+1)}{2} = \frac{1}{2}\binom{n}{2}\left(\binom{n}{2} + 1\right).$$

(c) A Riemann tensor has the pair antisymmetries, the pair-exchange symmetry, and the first Bianchi identity $R_{a[bcd]} = 0$. From part (b), before imposing Bianchi, the number is
$$\frac{N(N+1)}{2}, \qquad N = \binom{n}{2}.$$
The Bianchi identity removes the independent totally antisymmetric four-index part, whose dimension is $\binom{n}{4}$. Therefore
$$\#R = \frac{N(N+1)}{2} - \binom{n}{4} = \frac{n^2(n^2-1)}{12}.$$
For $n = 4$, this is
$$\frac{16(15)}{12} = 20.$$

Problem 15. Divergence

Problem. Prove the basic divergence identities for the Levi–Civita connection:

$$\nabla_\alpha \delta^\beta{}_\gamma = 0, \qquad \nabla_\mu U^\mu = \frac{1}{\sqrt{|g|}}\partial_\mu\left(\sqrt{|g|}U^\mu\right),$$

derive the corresponding formula for antisymmetric tensors, and show the coordinate expression for $\Box_g f$.

Solution.

(a) In coordinates,

$$\nabla_\alpha \delta^\beta{}_\gamma = \partial_\alpha \delta^\beta{}_\gamma + \Gamma^\beta{}_{\lambda\alpha}\delta^\lambda{}_\gamma - \Gamma^\lambda{}_{\gamma\alpha}\delta^\beta{}_\lambda = 0 + \Gamma^\beta{}_{\gamma\alpha} - \Gamma^\beta{}_{\gamma\alpha} = 0.$$

For the Levi–Civita connection, metric compatibility gives directly

$$\nabla_\alpha g_{\beta\gamma} = 0.$$

(b) Let $g = \det(g_{\alpha\beta})$. The derivative of the determinant with respect to a matrix entry is the corresponding cofactor:

$$\frac{\partial \det g}{\partial g_{\alpha\beta}} = \Delta^{\alpha\beta}.$$

Since $\Delta^{\alpha\beta} = g\, g^{\alpha\beta}$,

$$\partial_\mu g = g\, g^{\alpha\beta}\partial_\mu g_{\alpha\beta},$$

and therefore

$$g^{\alpha\beta}\partial_\mu g_{\alpha\beta} = \partial_\mu \log|g|.$$

For the Levi–Civita connection,

$$\Gamma^\alpha{}_{\mu\alpha} = \frac{1}{2}g^{\alpha\beta}\partial_\mu g_{\alpha\beta} = \partial_\mu \log\sqrt{|g|} = \frac{1}{\sqrt{|g|}}\partial_\mu\sqrt{|g|}.$$

Metric compatibility for $g^{\mu\alpha}$ gives

$$0 = \nabla_\mu g^{\mu\alpha} = \partial_\mu g^{\mu\alpha} + \Gamma^\mu{}_{\mu\lambda}g^{\lambda\alpha} + \Gamma^\alpha{}_{\mu\lambda}g^{\mu\lambda}.$$

Thus

$$g^{\mu\nu}\Gamma^\alpha{}_{\mu\nu} = -\frac{1}{\sqrt{|g|}}\partial_\mu\left(\sqrt{|g|}g^{\mu\alpha}\right).$$

For a scalar f,

$$\Box_g f = g^{\mu\nu}\nabla_\mu\nabla_\nu f = g^{\mu\nu}\partial_\mu\partial_\nu f - g^{\mu\nu}\Gamma^\alpha{}_{\mu\nu}\partial_\alpha f,$$

which becomes

$$\Box_g f = \frac{1}{\sqrt{|g|}}\partial_\mu\left(\sqrt{|g|}g^{\mu\nu}\partial_\nu f\right).$$

(c) For a vector field,

$$\nabla_\mu U^\mu = \partial_\mu U^\mu + \Gamma^\mu{}_{\mu\lambda}U^\lambda = \partial_\mu U^\mu + \frac{1}{\sqrt{|g|}}(\partial_\lambda\sqrt{|g|})U^\lambda,$$

hence

$$\nabla_\mu U^\mu = \frac{1}{\sqrt{|g|}}\partial_\mu\left(\sqrt{|g|}U^\mu\right).$$

If $F^{\mu\nu} = -F^{\nu\mu}$, then

$$\nabla_\mu F^{\mu\nu} = \partial_\mu F^{\mu\nu} + \Gamma^\mu{}_{\mu\lambda} F^{\lambda\nu} + \Gamma^\nu{}_{\mu\lambda} F^{\mu\lambda}.$$

The last term vanishes because $\Gamma^\nu{}_{\mu\lambda}$ is symmetric in μ, λ, while $F^{\mu\lambda}$ is antisymmetric. Therefore

$$\nabla_\mu F^{\mu\nu} = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} F^{\mu\nu} \right).$$

The same argument works for any totally antisymmetric contravariant tensor $A^{\mu\nu_1 \cdots \nu_k}$: every connection term on the remaining indices is a contraction of a symmetric Christoffel symbol with an antisymmetric pair, and therefore vanishes.

Problem 16. Parallel transport I

Problem. On the unit sphere with metric

$$h = \mathrm{d}\theta^2 + \sin^2\theta \, \mathrm{d}\phi^2,$$

start with $V = \partial_\theta$ at $(\theta, \phi) = (\theta_0, 0)$. Compute the change after parallel transport around the circle $\theta = \theta_0$. What is the magnitude of V?

Solution. Along the curve $\theta = \theta_0$, use ϕ as parameter. The non-zero Christoffel symbols are

$$\Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^\phi{}_{\theta\phi} = \Gamma^\phi{}_{\phi\theta} = \cot\theta.$$

Parallel transport gives

$$\frac{\mathrm{d}V^\theta}{\mathrm{d}\phi} - \sin\theta_0\cos\theta_0 \, V^\phi = 0, \qquad \frac{\mathrm{d}V^\phi}{\mathrm{d}\phi} + \cot\theta_0 \, V^\theta = 0.$$

It is convenient to use the orthonormal components

$$W^1 = V^\theta, \qquad W^2 = \sin\theta_0 \, V^\phi.$$

Then

$$\frac{\mathrm{d}W^1}{\mathrm{d}\phi} = \cos\theta_0 \, W^2, \qquad \frac{\mathrm{d}W^2}{\mathrm{d}\phi} = -\cos\theta_0 \, W^1.$$

With initial data $W^1(0) = 1,\ W^2(0) = 0$,

$$W^1(\phi) = \cos(\phi\cos\theta_0), \qquad W^2(\phi) = -\sin(\phi\cos\theta_0).$$

After one loop,

$$V^\theta(2\pi) = \cos(2\pi\cos\theta_0), \qquad V^\phi(2\pi) = -\frac{\sin(2\pi\cos\theta_0)}{\sin\theta_0}.$$

Thus the vector has been rotated, in the tangent plane, by angle $-2\pi\cos\theta_0$ relative to the transported orthonormal frame. Equivalently, relative to the original local frame, the holonomy angle is

$$2\pi(1 - \cos\theta_0)$$

modulo 2π, equal to the area enclosed by the latitude circle. Since parallel transport preserves the metric, the magnitude remains

$$|V| = 1.$$

Problem 17. Parallel transport II

Problem. For an infinitesimal loop spanned by two infinitesimal vectors X^μ and Y^μ, show that the change of a vector V^μ after parallel transport around the loop is

$$\delta V^\alpha = R^\alpha{}_{\beta\mu\nu} X^\mu Y^\nu V^\beta$$

up to the sign determined by the orientation convention for the loop.

Solution. Choose coordinates normal at the base point, so $\Gamma^\alpha{}_{\beta\mu} = 0$ at that point. To first non-trivial order, parallel transport around the parallelogram samples the first derivatives of the connection. Transport first along X, then Y, then $-X$, then $-Y$. Keeping only terms of order XY,

$$\delta V^\alpha = \left(\partial_\mu \Gamma^\alpha{}_{\beta\nu} - \partial_\nu \Gamma^\alpha{}_{\beta\mu}\right) X^\mu Y^\nu V^\beta.$$

In normal coordinates the quadratic $\Gamma\Gamma$ terms vanish at the base point, so this is exactly

$$\delta V^\alpha = R^\alpha{}_{\beta\mu\nu} X^\mu Y^\nu V^\beta.$$

Because both sides are tensorial at the base point, the identity holds in every coordinate system. Reversing the orientation of the loop changes the sign.

Problem 18. Killing vectors II

Problem. Show that a Killing vector K^μ satisfies

$$\nabla_\alpha \nabla^\alpha K_\beta + R_{\beta\sigma} K^\sigma = 0,$$

$$\nabla_\alpha \nabla_\beta K_\mu = R^\gamma{}_{\alpha\beta\mu} K_\gamma,$$

and

$$K^\sigma \nabla_\sigma R = 0.$$

Find a variational principle leading to the first equation.

Solution. The Killing equation is

$$\nabla_\alpha K_\beta + \nabla_\beta K_\alpha = 0.$$

Taking ∇^α gives

$$\nabla^\alpha \nabla_\alpha K_\beta + \nabla^\alpha \nabla_\beta K_\alpha = 0.$$

Since $\nabla^\alpha K_\alpha = 0$,

$$\nabla^\alpha \nabla_\beta K_\alpha = \left[\nabla^\alpha, \nabla_\beta\right] K_\alpha.$$

Using the curvature action on covectors,

$$\left[\nabla^\alpha, \nabla_\beta\right] K_\alpha = R_{\beta\sigma} K^\sigma,$$

with the sign convention used in this document. Therefore

$$\nabla_\alpha \nabla^\alpha K_\beta + R_{\beta\sigma} K^\sigma = 0.$$

For the second identity, differentiate the Killing equation and combine its three cyclic permutations:

$$\nabla_\alpha \nabla_\beta K_\mu \nabla_\alpha \nabla_\mu K_\beta = 0,$$

$$\nabla_\beta \nabla_\mu K_\alpha \nabla_\beta \nabla_\alpha K_\mu = 0,$$

$$\nabla_\mu \nabla_\alpha K_\beta \nabla_\mu \nabla_\beta K_\alpha = 0.$$

Adding the first two and subtracting the third, then commuting derivatives and using the Killing equation, yields

$$\nabla_\alpha \nabla_\beta K_\mu = R^\gamma{}_{\alpha\beta\mu} K_\gamma.$$

The flow of a Killing vector preserves the metric:

$$\mathcal{L}_K g_{\mu\nu} = \nabla_\mu K_\nu + \nabla_\nu K_\mu = 0.$$

Curvature scalars are natural functions of the metric and its Levi–Civita connection, hence they are also preserved by the flow:

$$\mathcal{L}_K R = K^\sigma \nabla_\sigma R = 0.$$

A Lagrangian whose Euler–Lagrange equation is the wave equation with Ricci coupling is

$$S[K] = \frac{1}{2} \int_M \left(\nabla_\alpha K_\beta \nabla^\alpha K^\beta - R_{\alpha\beta} K^\alpha K^\beta \right) \sqrt{|g|}\, \mathrm{d}^n x.$$

Varying K, integrating by parts, and discarding boundary terms gives

$$\nabla_\alpha \nabla^\alpha K_\beta + R_{\beta\sigma} K^\sigma = 0,$$

up to the global sign convention chosen for the action.

Problem 19. Killing vectors III

Problem. Show that a Killing vector, regarded as a vector potential, solves the source-free Maxwell equations in vacuum. Find the electromagnetic field corresponding to ∂_ϕ in Minkowski space.

Solution. Let $A_\mu = K_\mu$ and $F_{\mu\nu} = 2\nabla_{[\mu} K_{\nu]}$. Since $F = \mathrm{d}A$, the homogeneous Maxwell equation $\nabla_{[\alpha} F_{\mu\nu]} = 0$ holds identically. The inhomogeneous equation is

$$\nabla_\mu F^{\mu\nu} = \nabla_\mu (\nabla^\mu K^\nu - \nabla^\nu K^\mu) = \nabla_\mu \nabla^\mu K^\nu - R^\nu{}_\sigma K^\sigma.$$

In vacuum $R_{\mu\nu} = 0$, and the Killing-vector wave equation from Problem 18 gives $\nabla_\mu F^{\mu\nu} = 0$.

In cylindrical Minkowski coordinates,

$$g = -\mathrm{d}t^2 + \mathrm{d}\rho^2 + \rho^2 \mathrm{d}\phi^2 + \mathrm{d}z^2.$$

For $K = \partial_\phi$, $A = K_\mu \mathrm{d}x^\mu = \rho^2 \mathrm{d}\phi$. Hence

$$F = \mathrm{d}A = 2\rho\, \mathrm{d}\rho \wedge \mathrm{d}\phi.$$

In Cartesian coordinates this is

$$A = -y\, \mathrm{d}x + x\, \mathrm{d}y, \qquad F = 2\, \mathrm{d}x \wedge \mathrm{d}y,$$

which is a uniform magnetic field in the z-direction, up to the conventional normalisation of the vector potential.

Problem 20. Conformal transformation I

Problem. Show that a conformal transformation $\tilde{g} = f(x) g$ preserves angles and maps null curves to null curves.

Solution. For non-null vectors X, Y, the angle is determined by

$$\cos\theta = \frac{g(X,Y)}{\sqrt{|g(X,X)|}\sqrt{|g(Y,Y)|}}.$$

Under $\tilde{g} = fg$, assuming $f > 0$,

$$\frac{\tilde{g}(X,Y)}{\sqrt{|\tilde{g}(X,X)|}\sqrt{|\tilde{g}(Y,Y)|}} = \frac{fg(X,Y)}{\sqrt{|fg(X,X)|}\sqrt{|fg(Y,Y)|}} = \frac{g(X,Y)}{\sqrt{|g(X,X)|}\sqrt{|g(Y,Y)|}}.$$

Thus angles are preserved. If γ is null, then

$$\tilde{g}(\dot{\gamma},\dot{\gamma}) = f\,g(\dot{\gamma},\dot{\gamma}) = 0,$$

so the same curve is still null. Its affine parameter may change.

Problem 21. Conformal transformation II

Problem. Compute the curvature of $g_{\alpha\beta} = e^{2\varphi}\eta_{\alpha\beta}$, where η is flat.

Solution. Let n be the dimension and use η to raise indices in this formula. The Christoffel symbols are

$$\Gamma^{\alpha}{}_{\beta\gamma} = \delta^{\alpha}{}_{\beta}\partial_\gamma\varphi + \delta^{\alpha}{}_{\gamma}\partial_\beta\varphi - \eta_{\beta\gamma}\partial^\alpha\varphi.$$

The Ricci tensor is

$$R_{\mu\nu} = -(n-2)\partial_\mu\partial_\nu\varphi - \eta_{\mu\nu}\Box_\eta\varphi + (n-2)\left(\partial_\mu\varphi\partial_\nu\varphi - \eta_{\mu\nu}(\partial\varphi)^2\right).$$

The scalar curvature is

$$R = e^{-2\varphi}\left[-2(n-1)\Box_\eta\varphi - (n-1)(n-2)(\partial\varphi)^2\right].$$

The full Riemann tensor can be written compactly using the Schouten tensor

$$P_{\mu\nu} = -\partial_\mu\partial_\nu\varphi + \partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}\eta_{\mu\nu}(\partial\varphi)^2.$$

Because the metric is conformally flat,

$$R_{\alpha\beta\gamma\delta} = e^{2\varphi}\big(\eta_{\alpha\gamma}P_{\beta\delta} - \eta_{\alpha\delta}P_{\beta\gamma} - \eta_{\beta\gamma}P_{\alpha\delta} + \eta_{\beta\delta}P_{\alpha\gamma}\big).$$

For spacetime applications one usually sets $n = 4$.

Problem 22. Geodesics I

Problem. If a curve satisfies

$$\frac{D\dot{\gamma}}{\mathrm{d}\lambda} = \alpha(\lambda)\dot{\gamma},$$

show that it can be reparametrised to become affinely parametrised.

Solution. Let $\lambda = \lambda(\tau)$. Then

$$\frac{\mathrm{d}\gamma}{\mathrm{d}\tau} = \lambda'\dot{\gamma},$$

and

$$\frac{D}{\mathrm{d}\tau}\frac{\mathrm{d}\gamma}{\mathrm{d}\tau} = \lambda'^2\frac{D\dot{\gamma}}{\mathrm{d}\lambda} + \lambda''\dot{\gamma} = \left(\lambda'^2\alpha + \lambda''\right)\dot{\gamma}.$$

We need

$$\lambda'' + \alpha(\lambda)\lambda'^2 = 0.$$

Writing $p = \lambda'$, this is

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} + \alpha p^2 = 0, \qquad \frac{\mathrm{d}p}{\mathrm{d}\lambda}p + \alpha p^2 = 0,$$

so, for $p \neq 0$,

$$\frac{\mathrm{d}p}{\mathrm{d}\lambda} + \alpha p = 0.$$

Thus

$$p = C\exp\left(-\int^{\lambda}\alpha(\ell)\,\mathrm{d}\ell\right).$$

Solving $\mathrm{d}\lambda/\mathrm{d}\tau = p$ gives the required affine parameter.

Problem 23. Geodesics II

Problem. Show that the acceleration

$$a^\mu = \frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu{}_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda}$$

and, more generally,

$$\frac{DW^\mu}{\mathrm{d}\lambda} = \frac{\mathrm{d}W^\mu}{\mathrm{d}\lambda} + \Gamma^\mu{}_{\alpha\beta}W^\alpha\frac{\mathrm{d}\gamma^\beta}{\mathrm{d}\lambda}$$

transform as vectors.

Solution. For a vector W^μ,

$$\tilde{W}^\mu = \frac{\partial\tilde{x}^\mu}{\partial x^\nu}W^\nu.$$

Differentiate along γ:

$$\frac{\mathrm{d}\tilde{W}^\mu}{\mathrm{d}\lambda} = \frac{\partial\tilde{x}^\mu}{\partial x^\nu}\frac{\mathrm{d}W^\nu}{\mathrm{d}\lambda} + \frac{\partial^2\tilde{x}^\mu}{\partial x^\rho\partial x^\nu}\dot{x}^\rho W^\nu.$$

Adding the transformed Christoffel term and using the inhomogeneous transformation law of Γ, the second-derivative terms cancel exactly. The result is

$$\frac{\tilde{D}W^\mu}{\mathrm{d}\lambda} = \frac{\partial\tilde{x}^\mu}{\partial x^\nu}\frac{DW^\nu}{\mathrm{d}\lambda}.$$

Taking $W^\mu = \dot{x}^\mu$ gives the transformation law for a^μ.

Problem 24. Geodesics III

Problem. If a geodesic is timelike, spacelike, or null at one point, show that it has the same causal character everywhere.

Solution. For an affinely parametrised geodesic with tangent u,

$$\frac{\mathrm{d}}{\mathrm{d}s}g(u,u) = 2g(\nabla_u u, u) = 0.$$

Thus $g(u,u)$ is constant along the geodesic. Its sign, and hence the causal character of the geodesic, cannot change.

Problem 25. Static observers from a timelike Killing field

Problem. Let $u^\mu = K^\mu/\sqrt{-K^\nu K_\nu}$, where K is timelike and Killing. Show that

$$a_\mu := u^\nu\nabla_\nu u_\mu = \frac{1}{2}\nabla_\mu\log(-K^\alpha K_\alpha).$$

Solution. Set $N = \sqrt{-K^2}$, so $u_\mu = K_\mu/N$. Then

$$a_\mu = \frac{K^\nu}{N}\nabla_\nu\left(\frac{K_\mu}{N}\right) = \frac{1}{N^2}K^\nu\nabla_\nu K_\mu - \frac{K_\mu}{N^3}K^\nu\nabla_\nu N.$$

Since K is Killing,

$$K^\nu\nabla_\nu K_\mu = -\frac{1}{2}\nabla_\mu K^2.$$

Also $K^\nu\nabla_\nu K^2 = 0$, so $K^\nu\nabla_\nu N = 0$. Therefore

$$a_\mu = -\frac{1}{2N^2}\nabla_\mu K^2 = \frac{1}{2}\nabla_\mu\log(-K^2).$$

Problem 26. Kinematic decomposition of ∇u

Problem. For a fluid four-velocity u, show that

$$\nabla_\mu u_\nu = \omega_{\nu\mu} + \sigma_{\nu\mu} + \frac{\theta}{3} P_{\nu\mu} - a_\nu u_\mu,$$

with the definitions on the sheet.

Solution. Assume $u^\mu u_\mu = -1$ and

$$P_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu.$$

Since $\nabla_\mu (u^\nu u_\nu) = 0$, one has

$$u^\nu \nabla_\mu u_\nu = 0,$$

so the second index of $\nabla_\mu u_\nu$ is spatial. Decompose the first index using

$$\delta^\alpha{}_\mu = P^\alpha{}_\mu - u^\alpha u_\mu.$$

Then

$$\nabla_\mu u_\nu = P^\alpha{}_\mu \nabla_\alpha u_\nu - u_\mu u^\alpha \nabla_\alpha u_\nu = P^\alpha{}_\mu \nabla_\alpha u_\nu - a_\nu u_\mu.$$

The spatial tensor $P^\alpha{}_\mu \nabla_\alpha u_\nu$ splits into its antisymmetric part, symmetric trace-free part, and trace:

$$P^\alpha{}_\mu \nabla_\alpha u_\nu = \omega_{\nu\mu} + \sigma_{\nu\mu} + \frac{1}{3}\theta P_{\nu\mu}.$$

This is the desired decomposition.

Problem 27. Raychaudhuri equation

Problem. Using Problem 26, derive

$$\frac{\mathrm{d}\theta}{\mathrm{d}\tau} = \nabla_\mu a^\mu + 2(\omega^2 - \sigma^2) - \frac{\theta^2}{3} - R_{\mu\nu} u^\mu u^\nu.$$

Solution. Let $B_{\mu\nu} = \nabla_\nu u_\mu$. Along the flow,

$$u^\alpha \nabla_\alpha \theta = u^\alpha \nabla_\alpha \nabla_\mu u^\mu.$$

Commuting derivatives gives

$$u^\alpha \nabla_\alpha \nabla_\mu u^\mu = \nabla_\mu (u^\alpha \nabla_\alpha u^\mu) - (\nabla_\mu u^\alpha)(\nabla_\alpha u^\mu) - R_{\mu\nu} u^\mu u^\nu.$$

Thus

$$\frac{\mathrm{d}\theta}{\mathrm{d}\tau} = \nabla_\mu a^\mu - B_{\alpha\mu} B^{\mu\alpha} - R_{\mu\nu} u^\mu u^\nu.$$

Using

$$B_{\mu\nu} = \omega_{\mu\nu} + \sigma_{\mu\nu} + \frac{1}{3}\theta P_{\mu\nu} - a_\mu u_\nu,$$

orthogonality and symmetry properties imply

$$B_{\alpha\mu} B^{\mu\alpha} = -2\omega^2 + 2\sigma^2 + \frac{1}{3}\theta^2.$$

Substitution gives the stated Raychaudhuri equation.

Problem 28. Units I

Problem. In geometrized units $G = c = k_B = 1$, express selected physical constants in centimeters.

Solution. With $G = c = k_B = 1$, mass, time, temperature, charge, and energy are all converted to powers of length. The useful conversion rules are

$$M_{\rm geom} = \frac{GM_{\rm SI}}{c^2}, \qquad t_{\rm geom} = ct_{\rm SI}, \qquad T_{\rm geom} = \frac{Gk_B T_{\rm SI}}{c^4}.$$

Using standard constants,

$$\hbar_{\rm geom} = \frac{G\hbar_{\rm SI}}{c^3} = \ell_P^2 \approx 2.612 \times 10^{-66}\ {\rm cm}^2.$$

For electric charge in Gaussian geometrized units,

$$e_{\rm geom} = \frac{\sqrt{G}\, e_{\rm esu}}{c^2} \approx 1.38 \times 10^{-34}\ {\rm cm}.$$

For the electron,

$$\frac{e}{m_e} \approx \frac{1.38 \times 10^{-34}}{Gm_e/c^2} \approx 2.04 \times 10^{21}.$$

The solar mass is

$$M_\odot = \frac{GM_\odot^{\rm SI}}{c^2} \approx 1.477 \times 10^5\ {\rm cm}.$$

The solar luminosity, an energy per unit time, is dimensionless in these units after multiplication by G/c^5:

$$L_\odot \approx \frac{G(3.828 \times 10^{26}{\rm W})}{c^5} \approx 1.05 \times 10^{-21}.$$

A temperature 300 K corresponds to

$$T_{\rm geom} = \frac{Gk_B(300)}{c^4} \approx 3.42 \times 10^{-62}\ {\rm cm}.$$

One year is

$$c(1\,{\rm yr}) \approx 9.46 \times 10^{17}\ {\rm cm}.$$

One volt has dimensions of electric potential, hence charge over length in Gaussian geometrized units; numerically one may convert through one electron-volt:

$$1\,{\rm eV} = \frac{G(1.602 \times 10^{-19}{\rm J})}{c^4} \approx 1.32 \times 10^{-58}\ {\rm cm},$$

so $1\,{\rm V} = (1\,{\rm eV})/e_{\rm geom} \approx 9.6 \times 10^{-25}$ in geometrized potential units.

Problem 29. Units II

Problem. Find natural units of mass, length, and time from $\hbar, G, c$.

Solution. Dimensional analysis gives the Planck units:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}}, \qquad m_P = \sqrt{\frac{\hbar c}{G}}.$$

They are the unique monomials in $\hbar, G, c$ with dimensions of length, time, and mass.

Problem 30. Diagonal metrics

Problem. Show that for a diagonal metric,

$$\Gamma^a{}_{aa} = \partial_a \log \sqrt{|g_{aa}|}, \quad \Gamma^a{}_{ab} = \partial_b \log \sqrt{|g_{aa}|}, \quad \Gamma^a{}_{bb} = -\frac{1}{2g_{aa}}\partial_a g_{bb}, \quad \Gamma^a{}_{bc} = 0,$$

where a, b, c are distinct as appropriate and no repeated-index summation is understood.

Solution. For a diagonal metric, $g^{aa} = 1/g_{aa}$ and $g_{ab} = 0$ for $a \neq b$. The definition

$$\Gamma^a{}_{bc} = \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc})$$

has only the $d = a$ term. Thus

$$\Gamma^a{}_{aa} = \frac{1}{2}g^{aa}\partial_a g_{aa} = \partial_a \log \sqrt{|g_{aa}|},$$

$$\Gamma^a{}_{ab} = \frac{1}{2}g^{aa}\partial_b g_{aa} = \partial_b \log \sqrt{|g_{aa}|},$$

$$\Gamma^a{}_{bb} = -\frac{1}{2}g^{aa}\partial_a g_{bb} = -\frac{1}{2g_{aa}}\partial_a g_{bb}.$$

If a, b, c are all distinct, every metric component appearing in the formula is off-diagonal, hence zero, so $\Gamma^a{}_{bc} = 0$.

Problem 31. A naked singularity

Problem. For Schwarzschild with $m < 0$,

$$g = -\left(1 - \frac{2m}{r}\right)\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1 - 2m/r} + r^2\mathrm{d}\Omega^2,$$

consider radial curves satisfying

$$\frac{\mathrm{d}t}{\mathrm{d}r} = \frac{1}{1 - 2m/r}, \qquad t(r) \to t_0 \quad (r \to 0).$$

Solve the equation, show the curve is null and geodesic, and compute the needed $\Gamma^\mu{}_{\alpha\beta}$ for $\alpha, \beta \in \{t, r\}$.

Solution. Let $F = 1 - 2m/r$. Then

$$\frac{\mathrm{d}t}{\mathrm{d}r} = \frac{1}{F} = \frac{r}{r - 2m}.$$

Integrating,

$$t(r) = r + 2m \log|r - 2m| + C.$$

For $m < 0$, $r - 2m > 0$ for all $r > 0$, and C can be chosen so that

$$t(r) = t_0 + r + 2m \log\left(\frac{r - 2m}{-2m}\right).$$

This is defined for every $r > 0$. For $m > 0$, the usual Schwarzschild coordinate expression has a logarithmic obstruction at $r = 2m$.

The tangent with parameter r is

$$\dot{\gamma} = \frac{1}{F}\partial_t + \partial_r.$$

Its norm is

$$g(\dot{\gamma}, \dot{\gamma}) = -F\frac{1}{F^2} + \frac{1}{F} = 0,$$

so the curve is null.

For the (t, r)-sector,

$$g_{tt} = -F, \qquad g_{rr} = F^{-1}, \qquad F' = \frac{2m}{r^2}.$$

The non-zero Christoffel symbols with t, r indices are

$$\Gamma^t{}_{tr} = \Gamma^t{}_{rt} = \frac{F'}{2F} = \frac{m}{r^2 F},$$

$$\Gamma^r{}_{tt} = \frac{1}{2}FF' = \frac{mF}{r^2}, \qquad \Gamma^r{}_{rr} = -\frac{F'}{2F} = -\frac{m}{r^2 F}.$$

The geodesic equation with r as parameter may be non-affine, but direct substitution gives

$$\nabla_{\dot{\gamma}}\dot{\gamma} = -\frac{F'}{F}\dot{\gamma},$$

so the curve is a null geodesic after an affine reparametrisation.

Problem 32. Isotropic coordinates

Problem. Transform a spherically symmetric metric

$$g = -e^{2A(r)}\mathrm{d}t^2 + e^{2B(r)}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$$

to isotropic form. Apply this to Schwarzschild.

Solution. In isotropic radius $\bar{r}$, the spatial metric should be

$$\Psi(\bar{r})^4 \left(\mathrm{d}\bar{r}^2 + \bar{r}^2\mathrm{d}\Omega^2\right)$$

in three spatial dimensions. Comparing angular terms gives

$$r^2 = \Psi^4\bar{r}^2, \qquad \Psi^2 = \frac{r}{\bar{r}}.$$

Comparing radial terms gives

$$e^{2B(r)}\mathrm{d}r^2 = \Psi^4\mathrm{d}\bar{r}^2 = \frac{r^2}{\bar{r}^2}\mathrm{d}\bar{r}^2.$$

Thus

$$\frac{\mathrm{d}\bar{r}}{\bar{r}} = e^{B(r)}\frac{\mathrm{d}r}{r}.$$

This determines $\bar{r}$ up to a constant scale.

For Schwarzschild, $e^{2B} = 1/(1 - 2m/r)$, and integration yields

$$r = \bar{r}\left(1 + \frac{m}{2\bar{r}}\right)^2.$$

Therefore

$$g = -\left(\frac{1 - m/(2\bar{r})}{1 + m/(2\bar{r})}\right)^2 \mathrm{d}t^2 + \left(1 + \frac{m}{2\bar{r}}\right)^4 \left(\mathrm{d}\bar{r}^2 + \bar{r}^2\mathrm{d}\Omega^2\right).$$

Problem 33. Integral curves

Problem. Find integral curves of several vector fields on $\mathbb{R}^2$. Then show that if $g(\nabla f, \nabla f) = \psi(f)$, integral curves of ∇f are geodesics after reparametrisation. Apply this to the Schwarzschild and Eddington–Finkelstein coordinates described in the sheet.

Solution. For $X = \partial_x$, integral curves satisfy

$$\dot{x} = 1, \qquad \dot{y} = 0,$$

so $x = \lambda + x_0,\ y = y_0$. For $X = x\partial_y + y\partial_x$,

$$\dot{x} = y, \qquad \dot{y} = x,$$

hence

$$x = Ae^{\lambda} + Be^{-\lambda}, \qquad y = Ae^{\lambda} - Be^{-\lambda}.$$

For $X = x\partial_y - y\partial_x$,

$$\dot{x} = -y, \qquad \dot{y} = x,$$

so the integral curves are circles,

$$x = R\cos(\lambda + \lambda_0), \qquad y = R\sin(\lambda + \lambda_0).$$

For $X = x\partial_x + y\partial_y$,

$$x = x_0 e^{\lambda}, \qquad y = y_0 e^{\lambda}.$$

Now suppose $g(\nabla f, \nabla f) = \psi(f)$. Taking ∇_α gives

$$\nabla_\alpha(\nabla_\mu f \nabla^\mu f) = 2\nabla^\mu f \nabla_\alpha \nabla_\mu f = \psi'(f)\nabla_\alpha f.$$

For a scalar and a torsion-free connection, $\nabla_\alpha\nabla_\mu f = \nabla_\mu\nabla_\alpha f$, so

$$\nabla^\mu f \nabla_\mu \nabla_\alpha f = \frac{1}{2}\psi'(f)\nabla_\alpha f.$$

If $\dot{\gamma}^\mu = \nabla^\mu f$, this says

$$\frac{D\dot{\gamma}_\alpha}{\mathrm{d}\lambda} = \frac{1}{2}\psi'(f)\dot{\gamma}_\alpha,$$

or, after raising an index,

$$\frac{D\dot{\gamma}^\mu}{\mathrm{d}\lambda} = \frac{1}{2}\psi'(f)\dot{\gamma}^\mu.$$

By Problem 22, choose $\lambda = \lambda(\tau)$ satisfying

$$\lambda'' + \frac{1}{2}\psi'(f(\lambda))(\lambda')^2 = 0$$

to obtain an affine geodesic parameter.

For Schwarzschild,

$$g = -F\mathrm{d}t^2 + F^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \qquad F = 1 - \frac{2m}{r}.$$

For $f = t$, $g(\nabla t, \nabla t) = g^{tt} = -F^{-1}$, a function of r, not of t. For $f = r$, however,

$$g(\nabla r, \nabla r) = g^{rr} = F = 1 - \frac{2m}{r},$$

a function of $r = f$. Thus integral curves of $\nabla r = F\partial_r$ at fixed t, θ, ϕ are spacelike radial geodesics outside the horizon, after reparametrisation.

In ingoing Eddington–Finkelstein coordinates,

$$g = -F\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2.$$

The inverse two-dimensional block has

$$g^{vv} = 0, \qquad g^{vr} = 1, \qquad g^{rr} = F.$$

For $f = v$,

$$g(\nabla v, \nabla v) = 0,$$

so the integral curves of $\nabla v = \partial_r$ are null geodesics. For $f = r$, $g(\nabla r, \nabla r) = F(r)$, so integral curves of

$$\nabla r = \partial_v + F\partial_r$$

are also geodesics after reparametrisation; those meeting $r = 2m$ are null at the horizon. The family $\nabla v = \partial_r$ is the ingoing radial null family already represented by constant v, θ, ϕ, while the ∇r-family is distinct except at the horizon where it becomes null.

Problem 34. White hole

Problem. Let $m > 0$. Show that the Eddington–Finkelstein extension using

$$u = t - r - 2m\ln\left(\frac{r}{2m} - 1\right)$$

extends the exterior by a white-hole region: future-directed causal curves can exit it but cannot enter it.

Solution. The coordinate $u = t - r_*$ gives the outgoing Eddington–Finkelstein metric

$$g = -F\,\mathrm{d}u^2 - 2\,\mathrm{d}u\,\mathrm{d}r + r^2\mathrm{d}\Omega^2, \qquad F = 1 - \frac{2m}{r}.$$

It is smooth at $r = 2m$. For a future-directed causal vector

$$X = X^u\partial_u + X^r\partial_r + \text{angular part},$$

causality implies

$$0 \geq g(X, X) = -F(X^u)^2 - 2X^uX^r + r^2|\Omega(X)|^2.$$

At $r = 2m$, this becomes

$$-2X^uX^r + r^2|\Omega(X)|^2 \leq 0.$$

For future-directed curves in this outgoing chart one has $X^u > 0$. Hence at the horizon

$$X^r \geq 0.$$

Thus future-directed causal curves can cross the horizon only toward increasing r, from $r < 2m$ to $r > 2m$. They can leave the $r < 2m$ region, but cannot enter it from the exterior. This is the defining causal behaviour of a white hole region.

Problem 35. Proper time to the singularity

Problem. Show that an observer crossing the Schwarzschild radius $r = 2M$ reaches $r = 0$ in proper time $\tau \le \pi M$, independently of initial velocity.

Solution. Inside the horizon, the maximum possible proper time from $r = 2M$ to $r = 0$ is realised by a radial timelike geodesic with zero angular momentum. Any angular motion only uses part of the timelike budget and decreases the available proper time. For radial timelike geodesics,

$$-1 = -F\dot{t}^2 + F^{-1}\dot{r}^2, \qquad E = F\dot{t},$$

so

$$\dot{r}^2 = E^2 - F = E^2 - 1 + \frac{2M}{r}.$$

The proper time from $2M$ to 0 is

$$\tau = \int_0^{2M} \frac{\mathrm{d}r}{\sqrt{E^2 - 1 + 2M/r}}.$$

The largest value occurs for $E = 0$, giving

$$\tau_{\max} = \int_0^{2M} \frac{\mathrm{d}r}{\sqrt{2M/r - 1}}.$$

Set $r = 2M\sin^2\chi$. Then

$$\mathrm{d}r = 4M\sin\chi\cos\chi\,\mathrm{d}\chi, \qquad \sqrt{\frac{2M}{r} - 1} = \cot\chi.$$

Therefore

$$\tau_{\max} = 4M\int_0^{\pi/2} \sin^2\chi\,\mathrm{d}\chi = \pi M.$$

Hence every observer crossing the horizon reaches $r = 0$ within proper time at most πM.

Problem 36. Penrose diagrams

Problem. Compactify Minkowski space and Kruskal–Schwarzschild using $\hat{u} = \tan u$, $\hat{v} = \tan v$, and draw the conformal diagrams. Show that the two-dimensional projection of a causal vector is causal for $-\mathrm{d}t^2 + \mathrm{d}r^2$.

Solution. In Minkowski space set

$$\hat{u} = t_{\text{old}} - r_{\text{old}}, \qquad \hat{v} = t_{\text{old}} + r_{\text{old}},$$

and then $\hat{u} = \tan u$, $\hat{v} = \tan v$. With

$$t = v + u, \qquad r = v - u,$$

up to the harmless factor 1/2 depending on convention, one has

$$-\mathrm{d}t_{\text{old}}^2 + \mathrm{d}r_{\text{old}}^2 = -\mathrm{d}\hat{u}\,\mathrm{d}\hat{v} = -\sec^2 u\,\sec^2 v\,\mathrm{d}u\,\mathrm{d}v.$$

Thus the metric is conformal to

$$-\mathrm{d}t^2 + \mathrm{d}r^2 + r_{\text{old}}^2\mathrm{d}\Omega^2$$

on the diamond

$$v, u \in (-\pi/2, \pi/2), \qquad r = v - u > 0.$$

Equivalently,

$$-\pi < t + r < \pi, \qquad -\pi < t - r < \pi, \qquad r > 0.$$

For Kruskal–Schwarzschild,

$$g = -\frac{32m^3 e^{-R/(2m)}}{R} \,\mathrm{d}\hat{u}\,\mathrm{d}\hat{v} + R^2 \mathrm{d}\Omega^2,$$

where R is determined by $\hat{u}\hat{v}$. The same compactification gives

$$g = \Omega^{-2}(-\mathrm{d}u\,\mathrm{d}v) + R^2 \mathrm{d}\Omega^2$$

with a positive conformal factor wherever $R > 0$. The map is a diffeomorphism on

$$\hat{u}, \hat{v} \in \mathbb{R}, \qquad R(\hat{u}\hat{v}) > 0,$$

mapped to the finite square $u, v \in (-\pi/2, \pi/2)$ with the singular hyperbolae $R = 0$ removed.

For both metrics the angular term is non-negative:

$$g(X, X) = g_{(2)}(X_{(2)}, X_{(2)}) + R^2 |\Omega(X)|^2.$$

If X is causal, $g(X, X) \leq 0$, so

$$g_{(2)}(X_{(2)}, X_{(2)}) \leq 0.$$

Thus the projection is causal in the two-dimensional conformal Minkowski metric. The diagrams are therefore the usual Minkowski diamond and the four-region Kruskal diagram with two exterior regions, a black-hole future interior, and a white-hole past interior.

Problem 37. Metric connection with non-zero torsion

Problem. Repeat the curvature symmetry analysis for a metric connection with torsion and show

$$R^{\delta}{}_{[\alpha\beta\gamma]} = \nabla_{[\alpha} T^{\delta}{}_{\beta\gamma]} - T^{\epsilon}{}_{[\alpha\beta} T^{\delta}{}_{\gamma]\epsilon}.$$

Solution. Metric compatibility still implies

$$R_{abcd} = -R_{bacd},$$

because applying the curvature commutator to g_{ab} still gives zero. However, torsion changes the identities involving antisymmetrised derivatives. For a scalar,

$$\nabla_\alpha \nabla_\beta f - \nabla_\beta \nabla_\alpha f = -T^{\delta}{}_{\alpha\beta} \nabla_\delta f.$$

Apply this to $\nabla_\gamma f$:

$$(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) \nabla_\gamma f = -R^{\delta}{}_{\gamma\alpha\beta} \nabla_\delta f - T^{\delta}{}_{\alpha\beta} \nabla_\delta \nabla_\gamma f.$$

Now cyclically antisymmetrise over α, β, γ, commute the remaining second derivative of f once more, and collect the coefficients of $\nabla_\delta f$. Since f is arbitrary, the coefficient must vanish, which gives

$$R^{\delta}{}_{[\alpha\beta\gamma]} = \nabla_{[\alpha} T^{\delta}{}_{\beta\gamma]} - T^{\epsilon}{}_{[\alpha\beta} T^{\delta}{}_{\gamma]\epsilon}.$$

Thus the usual first Bianchi identity is recovered only when $T = 0$.

Problem 38. Wormhole

Problem. For

$$g = -\mathrm{d}t^2 + \mathrm{d}r^2 + (b^2 + r^2)\mathrm{d}\Omega^2,$$

an observer starts at $r = R$ with $u^r = -U$, $U > 0$. Compute the proper time to reach $r = -R$.

Solution. For radial motion with fixed angles,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2.$$

The metric coefficients are independent of t, so $u^t = \mathrm{d}t/\mathrm{d}\tau = E$ is constant. Normalisation gives

$$-1 = -(u^t)^2 + (u^r)^2.$$

Initially $u^r = -U$, hence

$$E^2 = 1 + U^2.$$

The radial geodesic equation is simply $\mathrm{d}^2r/\mathrm{d}\tau^2 = 0$, because ${\Gamma^r}_{tt} = {\Gamma^r}_{rr} = 0$ and the angular velocities vanish. Thus

$$r(\tau) = R - U\tau.$$

The arrival condition $r = -R$ gives

$$\tau = \frac{2R}{U}.$$

Problem 39. Timelike geodesics in Schwarzschild

Problem. For equatorial timelike Schwarzschild geodesics, use

$$\dot{t} = \frac{E}{1 - 2m/r}, \qquad \dot{\phi} = \frac{J}{r^2}$$

and $\lambda = 1$. Derive the radial equation and analyse the cases stated in the sheet.

Solution. The normalisation $g(\dot{\gamma}, \dot{\gamma}) = -1$, written in the convention of the sheet, gives

$$E^2 - \dot{r}^2 - \left(1 - \frac{2m}{r}\right)\left(1 + \frac{J^2}{r^2}\right) = 0.$$

Equivalently,

$$\frac{E^2 - \dot{r}^2}{1 - 2m/r} - \frac{J^2}{r^2} = 1,$$

which is the displayed formula.

For $E = 1$, $J = 4m$,

$$\dot{r}^2 = \frac{2m}{r} - \frac{16m^2}{r^2} + \frac{32m^3}{r^3} = \frac{2m(r - 4m)^2}{r^3}.$$

Also

$$\frac{\mathrm{d}r}{\mathrm{d}\phi} = \frac{\dot{r}}{\dot{\phi}} = \epsilon\frac{r^2}{4m}\sqrt{\frac{2m}{r}}\frac{|r - 4m|}{r}.$$

Integrating gives

$$\frac{\sqrt{r} - 2\sqrt{m}}{\sqrt{r} + 2\sqrt{m}} = Ae^{\epsilon\phi/\sqrt{2}}.$$

If $A = 0$, then $r = 4m$, the unstable circular orbit. If $A = 1, \epsilon = -1$, then $r \to \infty$ at $\phi = 0$ and the particle spirals toward $r = 4m$ as $\phi \to \infty$. If $r(0) = 3m, \epsilon = -1$, then

$$A = \frac{\sqrt{3} - 2}{\sqrt{3} + 2} < 0,$$

and the orbit falls inward to the singularity after a finite change of ϕ.

For $E = 1, J = 0$,

$$\dot{r}^2 = \frac{2m}{r}.$$

For infall,

$$\frac{\mathrm{d}r}{\mathrm{d}s} = -\sqrt{\frac{2m}{r}},$$

so

$$r^{3/2} = r_0^{3/2} - \frac{3}{2}\sqrt{2m}\,(s - s_0).$$

Problem 40. Circular geodesics

Problem. Show that for each $r > 3m$ there is an equatorial future-directed timelike geodesic with constant r, and derive the displayed formulae for J, t, ϕ.

Solution. The effective potential for timelike equatorial motion is

$$V_{\text{eff}}(r) = \left(1 - \frac{2m}{r}\right)\left(1 + \frac{J^2}{r^2}\right).$$

Circular motion requires $\dot{r} = 0$ and $V'_{\text{eff}} = 0$. Differentiating,

$$\frac{2m}{r^2}\left(1 + \frac{J^2}{r^2}\right) - 2\left(1 - \frac{2m}{r}\right)\frac{J^2}{r^3} = 0,$$

so

$$J^2 = \frac{mr^2}{r - 3m}.$$

This is positive exactly for $r > 3m$. The energy is

$$E^2 = \frac{(r - 2m)^2}{r(r - 3m)}.$$

Therefore

$$\dot{\phi} = \frac{J}{r^2}, \qquad \dot{t} = \frac{E}{1 - 2m/r} = \frac{rE}{r - 2m} = \sqrt{\frac{r}{r - 3m}} = \frac{|J|}{\sqrt{mr}}.$$

Integrating gives

$$t - t_0 = \frac{|J|}{\sqrt{mr}}(s - s_0), \qquad \phi - \phi_0 = \frac{J}{r^2}(s - s_0),$$

and hence

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \pm\sqrt{\frac{m}{r^3}}.$$

Problem 41. Null geodesics in Schwarzschild

Problem. For equatorial null geodesics with $J \neq 0$, set $u = m/r$ and $p = \mathrm{d}u/\mathrm{d}\phi$. Derive

$$p^2 = 2u^3 - u^2 + \alpha^2, \qquad \frac{\mathrm{d}u}{\mathrm{d}\phi} = p, \quad \frac{\mathrm{d}p}{\mathrm{d}\phi} = 3u^2 - u.$$

Analyse the critical points and solve the case $\alpha = 0,\ u > 1/2$.

Solution. For null geodesics,

$$\dot{r}^2 = E^2 - \left(1 - \frac{2m}{r}\right)\frac{J^2}{r^2}.$$

Since $\dot{\phi} = J/r^2$,

$$\left(\frac{\mathrm{d}r}{\mathrm{d}\phi}\right)^2 = \frac{r^4}{J^2}E^2 - r^2\left(1 - \frac{2m}{r}\right).$$

With $u = m/r$, $\mathrm{d}r/\mathrm{d}\phi = -(m/u^2)p$, so

$$p^2 = 2u^3 - u^2 + \frac{m^2E^2}{J^2}.$$

Thus $\alpha = mE/J$. Differentiating,

$$2p\frac{\mathrm{d}p}{\mathrm{d}\phi} = (6u^2 - 2u)p,$$

so away from turning points, and then by continuity,

$$\frac{\mathrm{d}u}{\mathrm{d}\phi} = p, \qquad \frac{\mathrm{d}p}{\mathrm{d}\phi} = 3u^2 - u.$$

Critical points satisfy $p = 0$ and $u(3u - 1) = 0$, hence

$$(u, p) = (0, 0), \qquad (1/3, 0).$$

The point $u = 1/3$ is the photon sphere $r = 3m$ and is a saddle in the phase portrait; $u = 0$ is the asymptotic end.

If $\alpha = 0$ and $u > 1/2$,

$$p^2 = u^2(2u - 1).$$

The proposed curve

$$r = 2m\cos^2\left(\frac{\phi - \phi_0}{2}\right)$$

has

$$u = \frac{1}{1 + \cos(\phi - \phi_0)}.$$

Direct differentiation verifies $p^2 = 2u^3 - u^2$. For $t(\phi)$,

$$\frac{\mathrm{d}t}{\mathrm{d}\phi} = \frac{\dot{t}}{\dot{\phi}} = \frac{Er^2}{J(1 - 2m/r)} = \frac{\alpha r^2/m}{1 - 2m/r}.$$

Here $\alpha = 0$, so the conserved-energy parametrisation degenerates; the curve lies in the horizon/interior branch and is more naturally described in Kruskal or Eddington–Finkelstein coordinates. In Schwarzschild t, it is a constant-t limiting null curve only where the coordinate is not regular.

Problem 42. Weak-field light bending I

Problem. Using

$$\frac{m}{r} = (\alpha - \alpha^2)\frac{x}{r} + \alpha^2\left(1 + \frac{y^2}{r^2}\right),$$

find the asymptote $x = ay + b$ as $y \to \infty$, for small $\alpha = m/d$.

Solution. Let $x = ay + b + o(1)$ with $a = O(\alpha)$. Then

$$r = y\sqrt{1 + a^2} + ab + o(1), \qquad \frac{x}{r} = a + \frac{b}{y} + O(\alpha^2/y).$$

Also $y^2/r^2 = 1 + O(\alpha^2)$. Keeping terms through $O(\alpha^2)$,

$$\frac{m}{r} = \frac{m}{y} + O(y^{-2}) = (\alpha - \alpha^2)\left(a + \frac{b}{y}\right) + 2\alpha^2 + O(\alpha^3, y^{-2}).$$

The constant term must vanish:

$$(\alpha - \alpha^2)a + 2\alpha^2 = 0,$$

so

$$a = -\frac{2\alpha}{1-\alpha} = -2\alpha + O(\alpha^2).$$

The $1/y$ term gives

$$m = (\alpha - \alpha^2)b,$$

hence

$$b = \frac{m}{\alpha(1-\alpha)} = \frac{d}{1-\alpha} = d + m + O(\alpha^2 d).$$

Problem 43. Weak-field light bending II

Problem. Does $\delta\phi = 4mG/(dc^2)$ have coordinate-independent meaning?

Solution. The individual symbols r, d, and ϕ in the Schwarzschild coordinate derivation are coordinate-dependent. The invariant statement is made at null infinity in an asymptotically flat spacetime: compare the incoming and outgoing asymptotic directions of the same null geodesic using the asymptotic Minkowski structure. The scattering angle between these directions is invariant. In the weak-field limit, the Schwarzschild coordinate d at closest approach agrees with the impact parameter up to higher-order corrections, and the invariant scattering angle is

$$\delta\phi = \frac{4GM}{bc^2} + O\left(\frac{G^2M^2}{b^2c^4}\right).$$

Thus the leading formula is invariant when d is interpreted, to leading order, as the asymptotic impact parameter.

Problem 44. Radial geodesics in Schwarzschild

Problem. For a massive particle initially at rest in Schwarzschild, derive the radial fall formula in the sheet.

Solution. At $s = 0$, $U^r = U^\theta = U^\phi = 0$. Normalisation gives

$$-1 = -F(r_0)(U^t(0))^2,$$

so

$$U^t(0) = \frac{1}{\sqrt{1 - 2m/r_0}}.$$

The angular geodesic equations have the schematic form

$$\frac{\mathrm{d}U^\theta}{\mathrm{d}s} = F_\theta(U^\theta, U^\phi), \qquad \frac{\mathrm{d}U^\phi}{\mathrm{d}s} = F_\phi(U^\theta, U^\phi),$$

with no source term when $U^\theta = U^\phi = 0$. Uniqueness of ODEs gives

$$U^\theta = U^\phi = 0.$$

The conserved energy is

$$E = F(r)\frac{\mathrm{d}t}{\mathrm{d}s} = F(r_0)U^t(0) = \sqrt{1 - \frac{2m}{r_0}}.$$

Thus

$$\frac{\mathrm{d}t}{\mathrm{d}s} = \frac{\sqrt{1 - 2m/r_0}}{1 - 2m/r}.$$

Normalisation gives

$$-1 = -\frac{E^2}{F} + \frac{\dot{r}^2}{F},$$

so

$$\dot{r}^2 = E^2 - F = \left(1 - \frac{2m}{r_0}\right) - \left(1 - \frac{2m}{r}\right) = \frac{2m}{r} - \frac{2m}{r_0}.$$

For infall,

$$\frac{\mathrm{d}r}{\mathrm{d}s} = -\sqrt{\frac{2m}{r} - \frac{2m}{r_0}}.$$

Integrating with $r = r_0 \cos^2 \chi$ gives

$$s = \frac{r_0^{3/2}}{\sqrt{2m}}\left[\cos^{-1}\sqrt{\frac{r}{r_0}} + \sqrt{\frac{r}{r_0}\left(1 - \frac{r}{r_0}\right)}\right].$$

Problem 45. A non-physical black-hole metric

Problem. For

$$g = -\left(1 - \frac{a}{r}\right)^3 \mathrm{d}t^2 + \frac{\mathrm{d}r^2}{(1 - a/r)^3} + r^2\mathrm{d}\Omega^2,$$

construct a regular ingoing coordinate, discuss the black-hole region, stationary acceleration, redshift, and the singularity.

Solution. Let

$$F = \left(1 - \frac{a}{r}\right)^3.$$

Define an ingoing coordinate

$$v = t + \int \frac{\mathrm{d}r}{F}.$$

Then $\mathrm{d}t = \mathrm{d}v - \mathrm{d}r/F$, and

$$g = -F\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2.$$

This expression is smooth at $r = a$ and extends to $r > 0$. With the ingoing choice, $r < a$ is a black-hole region: future-directed causal curves inside cannot cross $r = a$ outward.

Stationary observers have

$$u = F^{-1/2}\partial_t.$$

For any static metric of this form,

$$a_\mu = \frac{1}{2}\nabla_\mu \log F.$$

Thus

$$a_r = \frac{1}{2}\frac{F'}{F} = \frac{3a}{2r^2(1 - a/r)}.$$

The gravitational redshift between stationary observers at r_e and r_o is

$$\frac{\nu_o}{\nu_e} = \sqrt{\frac{F(r_o)}{F(r_e)}} = \left(\frac{1 - a/r_o}{1 - a/r_e}\right)^{3/2}.$$

The given invariants

$$R = 2a^3r^{-5}, \qquad R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = 4a^2r^{-10}(\cdots)$$

diverge as $r \to 0$. Since scalar curvature invariants diverge, $r = 0$ is a true curvature singularity, not a coordinate singularity.

Problem 46. Twin problem in Schwarzschild

Problem. Alice moves freely on a circular orbit of radius R, while Bob is stationary at radius r_0. Find the condition for equal aging.

Solution. Bob's proper time is

$$\mathrm{d}\tau_B = \sqrt{1 - \frac{2m}{r_0}}\,\mathrm{d}t.$$

For a circular geodesic at radius R, Problem 40 gives

$$\frac{\mathrm{d}t}{\mathrm{d}\tau_A} = \sqrt{\frac{R}{R-3m}},$$

so

$$\mathrm{d}\tau_A = \sqrt{1 - \frac{3m}{R}}\,\mathrm{d}t.$$

Neglecting travel time between r_0 and R, equal aging requires

$$\sqrt{1 - \frac{3m}{R}} = \sqrt{1 - \frac{2m}{r_0}}.$$

Therefore

$$\frac{3m}{R} = \frac{2m}{r_0}, \qquad R = \frac{3}{2}r_0.$$

This is possible as a timelike circular geodesic only if $R > 3m$, which follows from $r_0 > 2m$.

Problem 47. Perihelion shift estimates

Problem. Estimate the dimensionless parameters and perihelion shifts for Earth, Mercury, and a short-period exoplanet.

Solution. For a nearly Keplerian orbit, the relativistic perihelion advance per orbit is

$$\Delta\phi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

In geometrised units this is $6\pi m/[a(1-e^2)]$. For Earth,

$$\frac{m_\odot}{a_\oplus} \approx 9.87 \times 10^{-9},$$

so

$$\Delta\phi_\oplus \approx 1.86 \times 10^{-7}\ \text{rad/orbit} \approx 0.0384''/\text{orbit} \approx 3.84''/\text{century}.$$

For Mercury, with $a = 5.79 \times 10^{10}$m and $e = 0.2056$,

$$\frac{m_\odot}{a} \approx 2.55 \times 10^{-8},$$

and

$$\Delta\phi_{\text{Merc}} \approx 5.02 \times 10^{-7}\ \text{rad/orbit} \approx 0.1035''/\text{orbit} \approx 43''/\text{century}.$$

For a circular one-day orbit around a $10M_\odot$ star, Kepler's law gives

$$a = \left(\frac{GM}{4\pi^2}P^2\right)^{1/3},$$

and hence

$$\Delta\phi = \frac{6\pi GM}{ac^2}.$$

This is much larger than in the Solar System because M/a is larger. The dimensionless angular-momentum combinations reduce, in Newtonian circular motion, to powers of

$$\frac{m}{r} = \frac{GM}{rc^2}, \qquad \frac{J_N^2}{(m_0 r)^2} \simeq \frac{GM}{r}, \qquad \frac{m^2 m_0^2}{J_N^2} \simeq \frac{m}{r},$$

after restoring the necessary powers of c to make them dimensionless.

Problem 48. Gyroscope on a circular Schwarzschild orbit

Problem. For a spin vector s parallel transported along a circular timelike geodesic in Schwarzschild, derive the displayed equations and solve for $s(t)$.

Solution. Along the circular orbit,

$$u^r = u^\theta = 0, \qquad \Omega = \frac{\mathrm{d}\phi}{\mathrm{d}t} = \sqrt{\frac{m}{r^3}}.$$

The equations $Ds^\mu/\mathrm{d}\tau = 0$, together with $g(u, s) = 0$, give

$$\frac{\mathrm{d}s^\theta}{\mathrm{d}t} = 0, \qquad s^t = r^2 u^\phi \left(1 - \frac{2m}{r}\right)^{-1} u^t s^\phi,$$

and

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} = (r - 3m)\Omega s^\phi, \qquad \frac{\mathrm{d}s^\phi}{\mathrm{d}t} = -\frac{\Omega}{r} s^r.$$

Thus

$$\frac{\mathrm{d}^2 s^r}{\mathrm{d}t^2} = -(1 - 3m/r)\Omega^2 s^r.$$

Let

$$\omega_p = \Omega\sqrt{1 - \frac{3m}{r}}.$$

Then

$$s^r = A\cos(\omega_p t) + B\sin(\omega_p t),$$

and

$$s^\phi = \frac{1}{(r - 3m)\Omega}\frac{\mathrm{d}s^r}{\mathrm{d}t}.$$

The θ-component is constant and s^t is then fixed by orthogonality. Relative to the orbital angle Ωt, this gives the geodetic precession.

Problem 49. Circular timelike geodesics

Problem. Show again that circular timelike geodesics exist for $r > 3m$, and derive

$$\phi = \phi_0 + \Omega t, \qquad \Omega = m^{1/2} r^{-3/2}.$$

Solution. This is the same effective-potential calculation as Problem 40. The condition $V'_{\text{eff}} = 0$ gives

$$J^2 = \frac{mr^2}{r - 3m},$$

so such geodesics exist exactly for $r > 3m$. Since

$$\dot{\phi} = \frac{J}{r^2}, \qquad \dot{t} = \frac{|J|}{\sqrt{mr}},$$

we get

$$\Omega = \frac{\dot{\phi}}{\dot{t}} = \pm\sqrt{\frac{m}{r^3}},$$

and hence

$$\phi = \phi_0 + \Omega t, \qquad t = t_0 \pm \frac{J}{\sqrt{mr}} s.$$

Problem 50. Linearised rotating metric

Problem. For

$$g = g_{\text{Schw}} - \frac{4J \sin^2\theta}{r} \mathrm{d}t\, \mathrm{d}\phi,$$

analyse axis geodesics and the gyroscope equation along them.

Solution. On the axis $\theta = 0$, $\sin\theta = 0$, so the cross term $g_{t\phi}$ vanishes. Its first θ-derivative is proportional to $\sin\theta\cos\theta$, also zero on the axis. Therefore the submanifold $\theta = 0$ is invariant under the geodesic equation: curves of the form

$$\gamma(\tau) = (t(\tau), r(\tau), 0, \phi_0)$$

are geodesics whenever the remaining t, r equations are satisfied.

For a spin vector s, $Ds^\mu/\mathrm{d}\tau = 0$. Along the axis the equation for s^r is homogeneous in s^r, so if $s^r = 0$ initially then uniqueness of linear ODEs gives $s^r = 0$ all along the curve.

With $s^r = 0$, the remaining spin components satisfy a linear system

$$\frac{\mathrm{d}s^A}{\mathrm{d}\tau} + \Gamma^A{}_{B\mu}\dot{x}^\mu s^B = 0, \qquad A, B \in \{t, \theta, \phi\},$$

evaluated on $\theta = 0$. The J-dependent part couples the transverse angular components and produces frame dragging. Explicitly, after taking the axis limit in a regular local Cartesian angular frame, the spin precesses with the Lense–Thirring angular velocity proportional to J/r^3.

Problem 51. Newtonian binding energy of a uniform star

Problem. Compute the gravitational potential energy of a spherical Newtonian star with constant density ρ and radius R.

Solution. The mass inside radius r is

$$m(r) = \frac{4}{3}\pi\rho r^3.$$

A shell of thickness dr has mass

$$\mathrm{d}m = 4\pi\rho r^2 \mathrm{d}r.$$

Bringing this shell from infinity into the field of the interior mass contributes

$$\mathrm{d}U = -\frac{m(r)\mathrm{d}m}{r} = -4\pi\rho r^2 \frac{m(r)}{r}\mathrm{d}r,$$

where $G = 1$. Thus

$$U = -\int_0^R \frac{m(r)}{r} 4\pi\rho r^2 \, \mathrm{d}r = -\int_0^R \frac{16\pi^2}{3}\rho^2 r^4 \, \mathrm{d}r = -\frac{16\pi^2}{15}\rho^2 R^5.$$

Since total mass $M = (4/3)\pi\rho R^3$,

$$U = -\frac{3}{5}\frac{M^2}{R},$$

or $U = -3GM^2/(5R)$ with G restored.

Problem 52. Lane–Emden equation

Problem. Derive the Lane–Emden equation for

$$\rho = \rho_c \theta^n, \qquad p = p_c \theta^{n+1},$$

solve $n = 0$ and $n = 1$, and verify the $n = 5$ solution.

Solution. Since

$$\rho = \rho_c \theta^n,$$

we have

$$\theta = \left(\frac{\rho}{\rho_c}\right)^{1/n},$$

and

$$p = p_c \theta^{n+1} = p_c \left(\frac{\rho}{\rho_c}\right)^{(n+1)/n} = K\rho^{(n+1)/n},$$

with

$$K = p_c \rho_c^{-(n+1)/n}.$$

Also

$$\frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}r} = K\frac{n+1}{n}\rho^{1/n-1}\frac{\mathrm{d}\rho}{\mathrm{d}r} = K(n+1)\rho_c^{1/n}\frac{\mathrm{d}\theta}{\mathrm{d}r}.$$

Substitution into hydrostatic equilibrium gives

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 K(n+1)\rho_c^{1/n-1}\frac{\mathrm{d}\theta}{\mathrm{d}r}\right) = -4\pi\rho_c\theta^n.$$

Equivalently,

$$\frac{K(n+1)}{4\pi}\rho_c^{1/n-1}\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\theta}{\mathrm{d}r}\right) = -\theta^n.$$

Let

$$r_n^2 = \frac{K(n+1)}{4\pi}\rho_c^{1/n-1}, \qquad r = r_n\xi.$$

Then

$$\frac{1}{\xi^2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\right) = -\theta^n.$$

For $n = 0$,

$$(\xi^2\theta')' = -\xi^2,$$

so regularity at 0 gives

$$\theta = 1 - \frac{\xi^2}{6}.$$

For $n = 1$, set $\theta = \psi/\xi$. Then

$$\psi'' = -\psi,$$

and regularity plus $\theta(0) = 1$ gives

$$\theta = \frac{\sin\xi}{\xi}.$$

For $n = 5$,

$$\theta = (1 + \xi^2/3)^{-1/2}.$$

A direct differentiation gives

$$\frac{1}{\xi^2}(\xi^2\theta')' = -(1 + \xi^2/3)^{-5/2} = -\theta^5,$$

as required.

Problem 53. A gravitational wave

Problem. For $g_{ab} = \eta_{ab} + f n_a n_b$, where n_a is constant null and $n^a\partial_a f = 0$, compute the connection and Ricci tensor, and find vacuum wave solutions.

Solution. Since $n^a n_a = 0$, the inverse metric is

$$g^{ab} = \eta^{ab} - f n^a n^b.$$

The $f n^a n^b$ term does not contribute to the Christoffel symbols because $n^a\partial_a f = 0$. Therefore

$$\Gamma^a{}_{bc} = \frac{1}{2}\eta^{ad}(n_d n_b \partial_c f + n_d n_c \partial_b f - n_b n_c \partial_d f).$$

The trace $\Gamma^d{}_{dc}$ vanishes. Moreover any contraction of n with Γ vanishes. Hence the quadratic Christoffel terms in the Ricci tensor vanish, and

$$R_{ab} = -\frac{1}{2} n_a n_b\, \eta^{cd}\partial_c\partial_d f$$

up to the sign convention for R_{ab}. Thus the vacuum equation is

$$\Box_\eta f = 0.$$

For

$$f = \alpha\sin(k_a x^a),$$

this requires $k^a k_a = 0$. The condition $n^a\partial_a f = 0$ gives

$$n^a k_a = 0.$$

In Minkowski space, two non-zero null covectors which are orthogonal are proportional. Thus k_a is proportional to n_a.

Problem 54. Small perturbations of Minkowski spacetime

Problem. For $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $h = O(\epsilon)$, derive the linearised inverse metric, Christoffel symbols, and Ricci tensor.

Solution. The inverse metric satisfies

$$(\eta_{\mu\alpha} + h_{\mu\alpha})(\eta^{\alpha\nu} + k^{\alpha\nu}) = \delta_\mu{}^\nu.$$

To first order,

$$k^{\mu\nu} = -h^{\mu\nu},$$

so

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + O(\epsilon^2).$$

Using η to raise and lower indices,

$$\Gamma^\alpha{}_{\beta\gamma} = \frac{1}{2}\eta^{\alpha\delta}(\partial_\beta h_{\delta\gamma} + \partial_\gamma h_{\delta\beta} - \partial_\delta h_{\beta\gamma}) + O(\epsilon^2),$$

or

$$\Gamma^\alpha{}_{\beta\gamma} = \frac{1}{2}(\partial_\beta h^\alpha{}_\gamma + \partial_\gamma h^\alpha{}_\beta - \partial^\alpha h_{\beta\gamma}) + O(\epsilon^2).$$

The quadratic Christoffel terms in Ricci are $O(\epsilon^2)$, hence

$$R_{\beta\delta} = \partial_\alpha \Gamma^\alpha{}_{\beta\delta} - \partial_\delta \Gamma^\alpha{}_{\beta\alpha} + O(\epsilon^2).$$

Substitution gives

$$R_{\beta\delta} = \frac{1}{2}\left[\partial_\alpha\partial_\beta h^\alpha{}_\delta + \partial_\alpha\partial_\delta h^\alpha{}_\beta - \Box_\eta h_{\beta\delta} - \partial_\beta\partial_\delta h^\alpha{}_\alpha\right] + O(\epsilon^2).$$

Problem 55. Geodesics in a linearised plane wave

Problem. For

$$g_{\mu\nu} = \eta_{\mu\nu} + A_{\mu\nu}\cos(k_\alpha x^\alpha),$$

in the transverse traceless gauge described in the sheet, introduce $u = t - z,\ v = t + z$ and derive the geodesic equations.

Solution. In TT gauge with propagation in the z-direction, the metric has the form

$$g = -\mathrm{d}u\,\mathrm{d}v + (\delta_{AB} + h_{AB}(u))\mathrm{d}x^A\mathrm{d}x^B, \qquad A, B \in \{x, y\},$$

where $h_{AB} = A_{AB}\cos(\omega u)$, $A^A{}_A = 0$. The geodesic Lagrangian is

$$L = -\dot{u}\dot{v} + \frac{1}{2}(\delta_{AB} + h_{AB}(u))\dot{x}^A\dot{x}^B,$$

up to an overall harmless factor. Since v, x, y are cyclic except through $h(u)$,

$$\dot{u} = C$$

is constant, and

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left((\delta_{AB} + h_{AB})\dot{x}^B\right) = 0.$$

The u-equation, or the normalisation condition, determines v.

If u is constant, then $\dot{u} = 0$; the transverse momenta are constant and the remaining equations integrate linearly. These are the first class of geodesics.

If u is not constant, use u as parameter. To first order in ϵ and for small spatial velocities,

$$\frac{\mathrm{d}^2 x^A}{\mathrm{d}u^2} = -\frac{\mathrm{d}h^A{}_B}{\mathrm{d}u}\frac{\mathrm{d}x^B}{\mathrm{d}u} + O(\epsilon^2).$$

Equivalently,

$$(\delta_{AB} + h_{AB})\frac{\mathrm{d}x^B}{\mathrm{d}u} = P_A,$$

where P_A is constant, so

$$\frac{\mathrm{d}x^A}{\mathrm{d}u} = P^A - h^A{}_B P^B + O(\epsilon^2).$$

Integrating once gives

$$x^A(u) = x_0^A + P^A(u - u_0) - P^B \int_{u_0}^{u} h^A{}_B(w)\,\mathrm{d}w + O(\epsilon^2).$$

The coordinate $v(u)$ follows from the normalisation

$$-\frac{\mathrm{d}v}{\mathrm{d}u} + (\delta_{AB} + h_{AB})\frac{\mathrm{d}x^A}{\mathrm{d}u}\frac{\mathrm{d}x^B}{\mathrm{d}u} = \kappa/C^2,$$

where $\kappa = 0, -1, +1$ for null, timelike, or spacelike convention.

Problem 56. LISA / freely falling mirrors

Problem. For two freely falling mirrors separated by coordinate distance L transverse to a linearised plane wave with $h_\times = 0$, compute the photon round-trip time to first order.

Solution. Take the wave metric in TT gauge,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}z^2 + (1 + h_+(t-z))\mathrm{d}x^2 + (1 - h_+(t-z))\mathrm{d}y^2.$$

For mirrors separated in the x-direction at fixed TT coordinates and with light travel time short compared with the wave period, treat h_+ as constant during one trip. Along a null ray with y, z constant,

$$0 = -\mathrm{d}t^2 + (1 + h_+)\mathrm{d}x^2,$$

so

$$\Delta t_{\text{one way}} = \sqrt{1 + h_+}\,L = L\left(1 + \frac{1}{2}h_+\right).$$

The round-trip coordinate time is therefore

$$\Delta t_{\text{round}} = 2L\left(1 + \frac{1}{2}h_+\right)$$

to first order, evaluated at the appropriate retarded time. This equals the instantaneous proper spatial distance calculation in TT gauge:

$$\ell = L\sqrt{1 + h_+} = L(1 + h_+/2).$$

For a separation in the y-direction, the sign of h_+ is reversed.

Problem 57. Unit normals and induced metrics

Problem. Find unit normals and induced metrics for the four hypersurfaces listed in the sheet.
Solution.

(a) For $S^2 \subset \mathbb{R}^3$ with radius $R = 1$,

$$n = \partial_r, \qquad \gamma = \mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2.$$

(b) In Schwarzschild at constant $t, r = R$,

$$n = \sqrt{1 - \frac{2m}{R}}\, \partial_r, \qquad \gamma = R^2(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2).$$

This is the outward spacelike unit normal inside a $t =$ constant slice.

(c) For $S = \{x^0 = \sqrt{1 + r^2}\}$ in Minkowski, write

$$F = x^0 - \sqrt{1 + r^2}.$$

Then

$$\partial_\mu F = (1, -x_i/\sqrt{1 + r^2}), \qquad g^{\mu\nu}\partial_\mu F \partial_\nu F = -\frac{1}{1 + r^2}.$$

The future unit normal is

$$n = \sqrt{1 + r^2}\, \partial_t + x^i \partial_i.$$

Using $t = f(x)$, the induced metric is

$$\gamma_{ij} = \delta_{ij} - \frac{x_i x_j}{1 + r^2}.$$

(d) For $S' = \{r = \sqrt{1 + t^2}\}$, use coordinates (t, θ, ϕ). Since

$$\mathrm{d}r = \frac{t}{\sqrt{1 + t^2}} \mathrm{d}t,$$

the induced metric is

$$\gamma = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2 = -\frac{1}{1 + t^2} \mathrm{d}t^2 + (1 + t^2)\mathrm{d}\Omega^2.$$

A unit normal is proportional to $\mathrm{d}F = \mathrm{d}r - t(1 + t^2)^{-1/2}\mathrm{d}t$, giving

$$n = t\, \partial_t + \sqrt{1 + t^2}\, \partial_r$$

with spacelike unit norm.

Problem 58. Flat slicing of Schwarzschild

Problem. Find $f(r)$ so that the hypersurface $t = f(r)$ in Schwarzschild has flat induced metric.
Solution. On $t = f(r)$,

$$\mathrm{d}t = f'(r)\mathrm{d}r.$$

The induced metric is

$$\gamma = \left(F^{-1} - F(f')^2\right) \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2, \qquad F = 1 - \frac{2m}{r}.$$

For this to be Euclidean polar space,

$$\gamma = \mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2,$$

we need

$$F^{-1} - F(f')^2 = 1.$$

Thus

$$F(f')^2 = F^{-1} - 1 = \frac{1-F}{F},$$

and

$$f'(r) = \pm \frac{\sqrt{1-F}}{F} = \pm \frac{\sqrt{2m/r}}{1-2m/r}.$$

Integrating gives the Painleve–Gullstrand slicing:

$$f(r) = \pm \left[2\sqrt{2mr} + 2m \log \left| \frac{\sqrt{r} - \sqrt{2m}}{\sqrt{r} + \sqrt{2m}} \right| \right] + \text{constant}.$$

Problem 59. Wave energy currents

Problem. For $\Box_g \phi = 0$ and

$$T_{\mu\nu} = \partial_\mu \phi \, \partial_\nu \phi - \frac{1}{2} (\nabla^\alpha \phi \partial_\alpha \phi) g_{\mu\nu},$$

show $\nabla_\mu T^{\mu\nu} = 0$, construct currents from Killing fields, and write energy/angular momentum on $t = 0$ slices.

Solution. A direct calculation gives

$$\nabla_\mu T^{\mu\nu} = (\Box_g \phi) \nabla^\nu \phi = 0.$$

If X is Killing and $J^\mu = T^{\mu\nu} X_\nu$, then

$$\nabla_\mu J^\mu = T^{\mu\nu} \nabla_\mu X_\nu = \frac{1}{2} T^{\mu\nu} (\nabla_\mu X_\nu + \nabla_\nu X_\mu) = 0.$$

Both ∂_t and ∂_ϕ are Killing in Minkowski and Schwarzschild because the metric components are independent of t and ϕ.

In Minkowski, on $t = 0$ with $n = \partial_t$,

$$E = \frac{1}{2} \int_{\mathbb{R}^3} \left((\partial_t \phi)^2 + |\nabla_{\mathbb{R}^3} \phi|^2 \right) \mathrm{d}^3 x.$$

In Schwarzschild, on a $t =$ constant slice,

$$E = \frac{1}{2} \int \left[F^{-1} (\partial_t \phi)^2 + F (\partial_r \phi)^2 + \frac{1}{r^2} |\nabla_{S^2} \phi|^2 \right] r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi,$$

up to the conventional lapse/volume placement; this is the conserved Killing energy. For $X = \partial_\phi$, the conserved angular momentum is

$$J = \int_S T_{\mu\nu} (\partial_\phi)^\nu n^\mu \, \mathrm{d}\Sigma,$$

which in Minkowski is

$$J = -\int_{\mathbb{R}^3} \partial_t \phi \, \partial_\phi \phi \, \mathrm{d}^3 x$$

with sign depending on the normal convention. The Schwarzschild expression is the same integrand with the Schwarzschild hypersurface volume form.

Problem 60. Quadrupole radiation from a Kepler orbit

Problem. For a point mass in a Kepler ellipse, derive the time-dependent quadrupole and the first-order-in-eccentricity expression for h_{xx}.

Solution. For a point mass m_0 in the $z = 0$ plane,

$$q_{ij} = m_0 x_i x_j = m_0 r(\phi(t))^2 \begin{pmatrix} \cos^2\phi & \sin\phi\cos\phi & 0 \\ \sin\phi\cos\phi & \sin^2\phi & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

For a Kepler orbit,

$$r(\phi) = \frac{p}{1 + e\cos\phi} = p(1 - e\cos\phi) + O(e^2).$$

To first order, Kepler motion gives

$$\phi(t) = \Omega t + 2e\sin\Omega t + O(e^2),$$

where $\Omega^2 = m/a^3$. Therefore

$$q_{xx} = m_0 r^2 \cos^2\phi = m_0 a^2 \left[\cos^2\Omega t - e\left(2\cos\Omega t\cos^2\Omega t + 2\sin^2\Omega t\cos\Omega t\right)\right] + O(e^2).$$

The bracket simplifies to

$$q_{xx} = m_0 a^2 \left[\frac{1}{2}(1 + \cos 2\Omega t) - 2e\cos\Omega t\right] + O(e^2)$$

up to time-independent trace terms. In the quadrupole formula,

$$h_{xx}^{\mathrm{TT}} \sim \frac{2G}{c^4 D}\frac{\mathrm{d}^2}{\mathrm{d}t^2} q_{xx}^{\mathrm{TT}}(t - D/c),$$

so the first-order eccentricity adds a harmonic at frequency Ω in addition to the circular 2Ω harmonic.

Problem 61. Cosmological redshift

Problem. A galaxy has redshift $z = 0.2$. With $H_0 = 72\,\mathrm{km\,s^{-1}Mpc^{-1}}$, how far away was it when the light was emitted?

Solution. For small redshift, Hubble's law gives

$$v \approx cz, \qquad d \approx \frac{v}{H_0} = \frac{cz}{H_0}.$$

Thus

$$d \approx \frac{(3.00 \times 10^5\mathrm{km/s})(0.2)}{72\mathrm{km\,s^{-1}Mpc^{-1}}} \approx 8.33 \times 10^2\mathrm{Mpc}.$$

This is the simple Hubble-law estimate; a precise FLRW answer would require the cosmological parameters.

Problem 62. Cepheid distance

Problem. A Cepheid has apparent magnitude $m = 22$ and period 28 days. Estimate its distance using the period–luminosity relation.

Solution. A typical period–luminosity calibration gives an absolute magnitude around

$$M \simeq -5$$

for a 28-day classical Cepheid. The distance modulus is

$$m - M = 5\log_{10}(d/10\text{pc}).$$

Thus

$$22 - (-5) = 27 = 5\log_{10}(d/10\text{pc}),$$

so

$$d = 10\text{pc} \times 10^{27/5} \approx 2.5 \times 10^6\text{pc} = 2.5\text{Mpc}.$$

Different calibrations shift the estimate modestly, but the distance is of order a few megaparsecs.

Problem 63. Null rays in FLRW

Problem. Show that radial null rays in FLRW are geodesics, perhaps non-affinely parametrised.

Solution. For

$$g = -\mathrm{d}t^2 + R(t)^2\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\Omega^2\right),$$

radial null curves satisfy

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \pm\frac{\sqrt{1-kr^2}}{R(t)}.$$

Equivalently, in conformal time $\mathrm{d}\eta = \mathrm{d}t/R(t)$,

$$g = R(\eta)^2\left[-\mathrm{d}\eta^2 + \frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\Omega^2\right].$$

Radial null curves are unchanged by the conformal factor. In the conformally related static metric they are ordinary radial null geodesics; therefore in the FLRW metric they are null geodesics, with affine parameter rescaled by the conformal factor.

Problem 64. Concavity of the FLRW scale factor

Problem. For $\rho + 3p \geq 0$ and $\Lambda \leq 0$, show $R''(t) \leq 0$. If $R(t) \sim Ct^\alpha$ as $t \to 0$, deduce $1/H(t) \leq t$ or $1/H(t) \geq t$ according to the convention in the sheet.

Solution. The acceleration Friedmann equation is

$$\frac{\ddot{R}}{R} = -\frac{4\pi}{3}(\rho + 3p) + \frac{\Lambda}{3}.$$

Under the assumptions, the right-hand side is non-positive, so

$$\ddot{R} \leq 0.$$

For an expanding universe, $H = \dot{R}/R$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{H}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{R}{\dot{R}}\right) = 1 - \frac{R\ddot{R}}{\dot{R}^2} \geq 1.$$

If $R(t) \sim Ct^\alpha$ near 0, then $1/H \sim t/\alpha$. Concavity with $R(0) = 0$ implies $0 < \alpha \leq 1$, hence $1/H \geq t$ near the big bang and, by the derivative inequality,

$$\frac{1}{H(t)} \geq t$$

for later times.

Problem 65. Matter-dominated Friedmann solutions

Problem. Derive the $k = \pm 1$, $\Lambda = 0$, matter-dominated parametric solutions.

Solution. For dust, $\rho R^3 =$ constant, and the Friedmann equation can be written

$$\dot{R}^2 = \frac{C_1}{R} - k.$$

For $k = 1$, set

$$R = C(1 - \cos\eta).$$

Then

$$\frac{\mathrm{d}R}{\mathrm{d}\eta} = C\sin\eta, \qquad \frac{\mathrm{d}t}{\mathrm{d}\eta} = R = C(1 - \cos\eta),$$

so

$$t = C(\eta - \sin\eta).$$

For $k = -1$, set

$$R = C(\cosh\eta - 1), \qquad \frac{\mathrm{d}t}{\mathrm{d}\eta} = R,$$

giving

$$t = C(\sinh\eta - \eta).$$

Substitution verifies both formulae.

Problem 66. Lifetime of a closed dust universe

Problem. A closed matter-dominated FLRW universe with $\Lambda = 0$ has spatial volume $10^{12}\mathrm{Mpc}^3$ at maximum expansion. Find its lifetime.

Solution. For a closed spherical spatial section,

$$V = 2\pi^2 R^3.$$

At maximum expansion,

$$R_{\max} = \left(\frac{10^{12}}{2\pi^2}\right)^{1/3}\mathrm{Mpc} \approx 3.70 \times 10^3\mathrm{Mpc}.$$

For the $k = 1$ dust solution,

$$R = C(1 - \cos\eta), \qquad t = C(\eta - \sin\eta).$$

The maximum occurs at $\eta = \pi$, so $R_{\max} = 2C$. The big crunch is at $\eta = 2\pi$, hence

$$t_{\text{life}} = 2\pi C = \pi R_{\max}.$$

Since $1\mathrm{Mpc}/c \approx 3.26 \times 10^6$ years,

$$t_{\text{life}} \approx \pi(3.70 \times 10^3)(3.26 \times 10^6) \approx 3.8 \times 10^{10} \text{ years}.$$

Problem 67. Instability at $\dot{R} = 0$

Problem. Show that all solutions of the linearised Friedmann equation for radiation and dust at $\dot{R} = 0$ are unstable.

Solution. Write the Friedmann equation as a mechanical energy equation

$$\dot{R}^2 + V(R) = 0.$$

For a mixture of dust and radiation,

$$\rho = \rho_m + \rho_r, \qquad \rho_m \propto R^{-3}, \quad \rho_r \propto R^{-4}.$$

Thus

$$\dot{R}^2 = \frac{A}{R} + \frac{B}{R^2} - k + \frac{\Lambda}{3} R^2.$$

At a static turning point $\dot{R} = 0$, linearising the second-order Friedmann equation gives

$$\delta \ddot{R} = \left[-\frac{4\pi}{3} \frac{\mathrm{d}}{\mathrm{d}R} \big((\rho + 3p) R \big) + \frac{\Lambda}{3} \right]_{R=R_0} \delta R.$$

For dust plus radiation this coefficient is positive at the Einstein-static balance point. Therefore perturbations contain exponential modes

$$\delta R \sim e^{\pm \omega t},$$

so every such static solution has an unstable mode.

Problem 68. Density scaling and equality redshift

Problem. Use the Friedmann and continuity equations to find $\rho(R)$ for radiation, matter, and cosmological constant. Then find the redshift of matter–radiation equality.

Solution. The continuity equation is

$$\dot{\rho} + 3H(\rho + p) = 0.$$

For an equation of state $p = w\rho$,

$$\frac{\dot{\rho}}{\rho} = -3(1 + w) \frac{\dot{R}}{R},$$

so

$$\rho \propto R^{-3(1+w)}.$$

Radiation has $w = 1/3$:

$$\rho_r \propto R^{-4}.$$

Matter has $w = 0$:

$$\rho_m \propto R^{-3}.$$

A cosmological constant has $w = -1$:

$$\rho_\Lambda = \text{constant}.$$

On a log–log plot of ρ versus R, these are straight lines with slopes -4, -3, and 0.

Since

$$\rho_m(z) = \rho_{m,0}(1 + z)^3, \qquad \rho_r(z) = \rho_{r,0}(1 + z)^4,$$

equality occurs when

$$1 + z_{\rm eq} = \frac{\rho_{m,0}}{\rho_{r,0}}.$$

Using the given values,

$$1 + z_{\rm eq} = \frac{1.88 \times 10^{-32} \Omega_m h^2}{7.8 \times 10^{-37}} \approx 2.41 \times 10^4 \Omega_m h^2.$$

Thus

$$z_{\rm eq} \approx 2.41 \times 10^4 \Omega_m h^2 - 1.$$